

# MC 1540 MC 1440

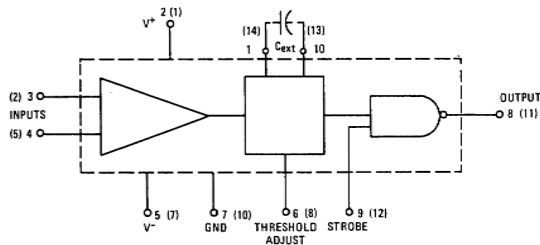
## SENSE AMPLIFIERS

### MONOLITHIC SENSE AMPLIFIER

... consisting of a wideband differential amplifier, a dc restoration circuit which also incorporates facilities to externally adjust the threshold, and an MDTL output gate which is strobed from saturated logic. It is designed to detect bipolar differential signals derived by a core memory with cycle times as low as 0.5  $\mu$ s.

- Differential Threshold Characteristics:  
Adjustable Threshold — 10–25 mV  
Nominal Threshold — 17 mV @  $V_G = -6$  V  
Input Offset Voltage — 1.0 mV typical  
Threshold Drift —  $-10 \mu$ V/ $^{\circ}$ C typical
- Fast Response Time — 20 ns typical
- Short Recovery Time:  
50 ns max @  $e_{in} = 1.8$  V Common Mode  
50 ns max @  $e_{in} = 400$  mV Differential Mode

### MC1540/MC1440 BLOCK DIAGRAM



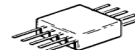
Number at end of terminal represents pin number for devices in flat package and metal can. Number in parenthesis represents pin number for dual in-line package.

### CORE MEMORY SENSE AMPLIFIER INTEGRATED CIRCUIT

SILICON  
EPITAXIAL PASSIVATED

F SUFFIX  
CERAMIC PACKAGE  
CASE 606  
TO-91

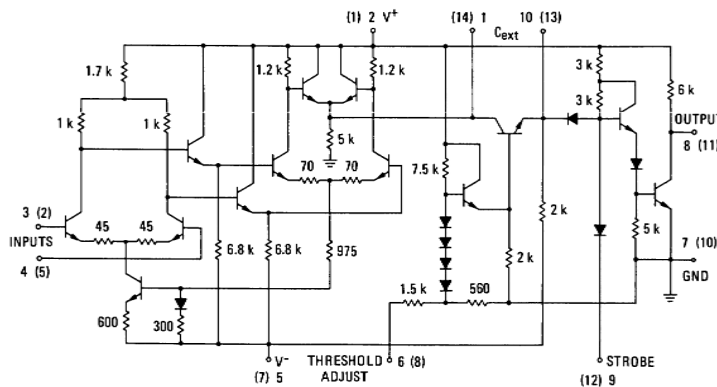
G SUFFIX  
METAL PACKAGE  
CASE 602B



L SUFFIX  
CERAMIC PACKAGE  
CASE 632  
TO-116



### CIRCUIT SCHEMATIC



Number at end of terminal represents pin number for devices in flat package and metal can. Number in parenthesis represents pin number for ceramic dual in-line package.

See Packaging Information Section for outline dimensions.

# MC1540, MC1440 (continued)

## MAXIMUM RATINGS (T<sub>A</sub> = +25°C unless otherwise noted)

| Rating                                 | Symbol                           | Value                   | Unit  |
|----------------------------------------|----------------------------------|-------------------------|-------|
| Power Supply Voltage                   | V <sup>+</sup><br>V <sup>-</sup> | +10<br>-10              | Vdc   |
| Differential Input Signal              | V <sub>in</sub>                  | ±5.0                    | Vdc   |
| Common Mode Input Voltage              | CMV <sub>in</sub>                | ±5.0                    | Vdc   |
| Load Current                           | I <sub>L</sub>                   | 25                      | mA    |
| Power Dissipation (Package Limitation) | P <sub>D</sub>                   |                         |       |
| Metal Can                              |                                  | 680                     | mW    |
| Derate above T <sub>A</sub> = +25°C    |                                  | 4.6                     | mW/°C |
| Flat Package                           |                                  | 500                     | mW    |
| Derate above T <sub>A</sub> = +25°C    |                                  | 3.3                     | mW/°C |
| Ceramic Dual In-Line Package           |                                  | 625                     | mW    |
| Derate above T <sub>A</sub> = +25°C    |                                  | 5.0                     | mW/°C |
| Operating Temperature Range            | T <sub>A</sub>                   | 0 to +75<br>-55 to +125 | °C    |
| Storage Temperature Range              | T <sub>stg</sub>                 | -65 to +150             | °C    |

## ELECTRICAL CHARACTERISTICS

(V<sup>+</sup> = +6 Vdc ± 1%, V<sup>-</sup> = -6 Vdc ± 1%, C<sub>ext</sub> = 0.01 μF, T<sub>A</sub> = +25°C unless otherwise noted)

Pin number references are for devices in flat package and metal can.

See block diagram for dual in-line package pin numbers.

| Characteristic                                                                                                                                                                                                                                                                                    | Fig. No.    | Symbol                                                       | MC1540         |                |                | MC1440         |                |                | Unit |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------------------------|----------------|----------------|----------------|----------------|----------------|----------------|------|
|                                                                                                                                                                                                                                                                                                   |             |                                                              | Min            | Typ            | Max            | Min            | Typ            | Max            |      |
| Input Threshold Voltage<br>(V <sub>G</sub> = -6.0 Vdc, T <sub>A</sub> = 25°C)<br>(V <sub>G</sub> = -6.0 V, T <sub>A</sub> = T <sub>low</sub> *)<br>(V <sub>G</sub> = -6.0 V, T <sub>A</sub> = T <sub>high</sub> *)                                                                                | 1           | V <sub>th</sub>                                              | 14<br>12<br>12 | 17<br>17<br>17 | 20<br>24<br>22 | 12<br>10<br>10 | 17<br>17<br>17 | 24<br>30<br>30 | mV   |
| Input Offset Voltage                                                                                                                                                                                                                                                                              | 1           | V <sub>io</sub>                                              | —              | 1.0            | 5.0            | —              | 1.0            | 6.0            | mV   |
| Input Bias Current<br>(V <sub>3</sub> = V <sub>4</sub> = 0, T <sub>A</sub> = 25°C)<br>(V <sub>3</sub> = V <sub>4</sub> = 0, T <sub>A</sub> = T <sub>low</sub> *)                                                                                                                                  | 2           | I <sub>b</sub>                                               | —<br>—         | 7.5<br>—       | 50<br>100      | —<br>—         | 7.5<br>—       | 75<br>100      | μA   |
| Input Offset Current                                                                                                                                                                                                                                                                              | 2           | I <sub>io</sub>                                              | —              | 2.0            | 10             | —              | 2.0            | 15             | μA   |
| Output Voltage High<br>(V <sub>3</sub> = V <sub>4</sub> = 0)                                                                                                                                                                                                                                      | 3           | V <sub>OH</sub>                                              | 5.9            | —              | —              | 5.8            | —              | —              | Vdc  |
| Output Voltage Low<br>(V <sub>3</sub> = V <sub>4</sub> = 0, V <sub>I0</sub> = +6.0 Vdc, I <sub>B</sub> = 6.0 mAdc)<br>(V <sub>I0</sub> = +6.0 Vdc, I <sub>B</sub> = 6.0 mAdc, T <sub>A</sub> = T <sub>high</sub> *)                                                                               | 3           | V <sub>OL</sub>                                              | —<br>—         | —<br>—         | 350<br>400     | —<br>—         | —<br>—         | 400<br>450     | mVdc |
| Amplifier Voltage Gain<br>(V <sub>3</sub> = 15 mV peak)                                                                                                                                                                                                                                           | 4           | A <sub>V</sub>                                               | —              | 85             | —              | —              | 85             | —              | —    |
| Strobe Load Current<br>(V <sub>G</sub> = 0)                                                                                                                                                                                                                                                       | —           | I <sub>S</sub>                                               | —              | —              | 1.2            | —              | —              | 1.5            | mAdc |
| Strobe Reverse Current<br>(V <sub>G</sub> = +5.0 Vdc)<br>(V <sub>G</sub> = +6.0 Vdc, T <sub>A</sub> = T <sub>high</sub> *)                                                                                                                                                                        | —           | I <sub>R</sub>                                               | —<br>—         | —<br>—         | 2.0<br>25      | —<br>—         | —<br>—         | 5.0<br>30      | μAdc |
| Propagation Delay<br>Input to Amplifier Output<br>(V <sub>3</sub> = 25 mV pulse, V <sub>G</sub> = +2.0 Vdc)<br>Input to Gate Output<br>(V <sub>3</sub> = 25 mV pulse, V <sub>G</sub> = +2.0 Vdc)<br>Strobe to Gate Output<br>(V <sub>3</sub> = V <sub>4</sub> = 0, V <sub>G</sub> = +2.0 V pulse) | 5<br>5<br>6 | t <sub>3+10+</sub><br>t <sub>3+8-</sub><br>t <sub>g+8-</sub> | —<br>—<br>—    | 10<br>20<br>10 | 15<br>30<br>15 | —<br>—<br>—    | 10<br>20<br>10 | 20<br>50<br>30 | ns   |
| Recovery Time<br>Differential Mode<br>(V <sub>3</sub> = 400 mV pulse)<br>Common Mode<br>(V <sub>3</sub> = 1.8 V pulse)                                                                                                                                                                            | 7<br>8      | t <sub>R(dm)</sub><br>t <sub>R(cm)</sub>                     | —<br>—         | 20<br>20       | 50<br>50       | —<br>—         | 20<br>20       | 90<br>60       | ns   |
| Power Dissipation                                                                                                                                                                                                                                                                                 | —           | P <sub>D</sub>                                               | —              | 120            | 180            | —              | 120            | 250            | mW   |

\*T<sub>low</sub> = -55°C for MC1540 or 0°C for MC1440, T<sub>high</sub> = +125°C for MC1540 or +75°C for MC1440.

- $A_V$  Amplifier Voltage Gain — the ratio of output voltage at pin 1 to the input voltage at pin 3 or 4
- $I_b$  Input Bias Current — the average input current defined as  $(I_3 + I_4)/2$
- $I_{io}$  Input Offset Current — the difference between input current values,  $|I_3 - I_4|$
- $I_R$  Strobe Reverse Current — leakage current when the strobe input is high
- $I_S$  Strobe Load Current — amount of current drain from the circuit when the strobe pin is grounded
- $P_D$  Power Dissipation — amount of power dissipated in the unit as defined by  $|I_2 \times V^+| + |I_5 \times V^-|$
- $t_R$  Recovery Time — The time that is required for the device to recover from the specified differential and common-mode overload inputs prior to strobe as reference to the 10% point

- of the trailing edge of an input pulse. The device is considered recovered when the threshold after a differential overload disturbance is within 1.0 mV of the threshold value without the disturbance, or, for common-mode disturbance, when the level at pin 10 is within 100 mV of the quiescent value.
- $t_{x \pm y \pm}$  Propagation Delay — The time that is required for the output pulse at pin y to achieve 50% of its final value or the 1.5 V level referenced to 50% of the input pulse at pin x. (The + and - denote positive and negative-going pulse transition.)
- $V_{OH}$  Output Voltage High — high-level output voltage when the output gate is turned off
- $V_{OL}$  Output Voltage Low — low-level output voltage when the output gate is turned on
- $V_{th}$  Input Threshold — input pulse amplitude that causes the output to begin saturation
- $V_{io}$  Input Offset Voltage — the difference in  $V_{th}$  at each input

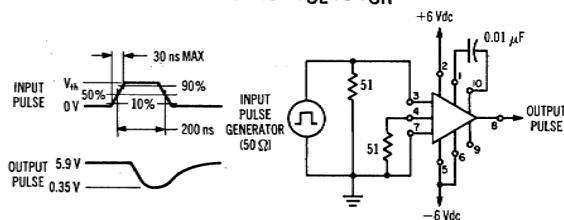
FIGURE 1 — INPUT THRESHOLD AT OUTPUT VOLTAGE SWING FROM  $V_{OL}$  TO  $V_{OH}$ 

FIGURE 3 — OUTPUT VOLTAGE LEVELS

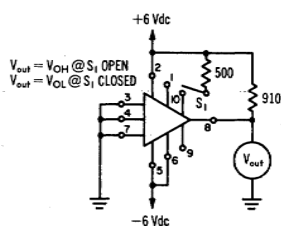


FIGURE 5 — PROPAGATION DELAY (STROBE HIGH)

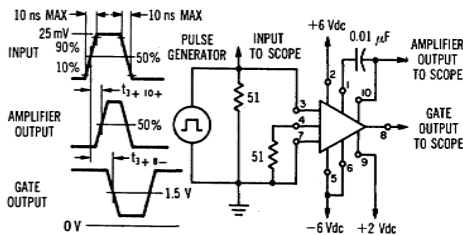


FIGURE 2 — INPUT BIAS CURRENT TEST CIRCUIT

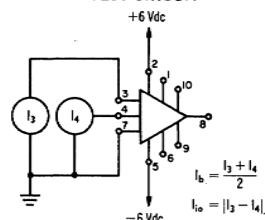


FIGURE 4 — AMPLIFIER VOLTAGE GAIN

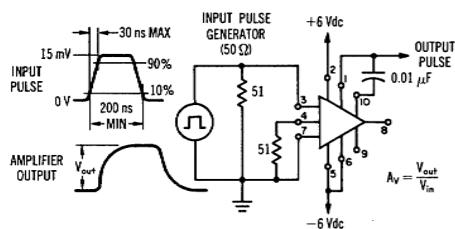
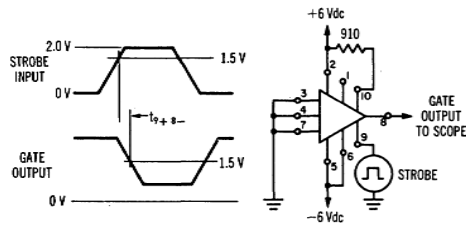
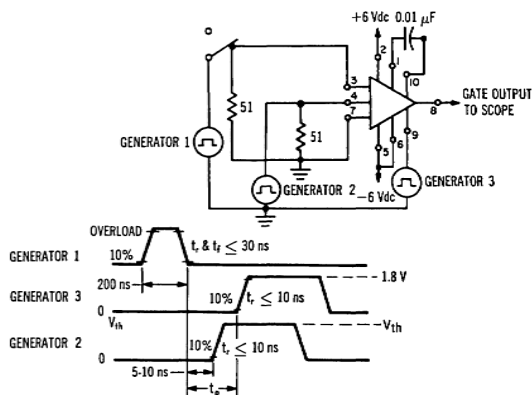


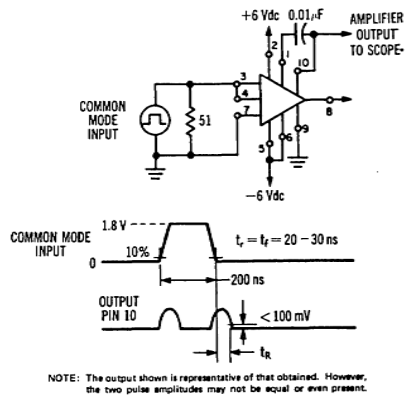
FIGURE 6 — PROPAGATION DELAY (STROBE INPUT)



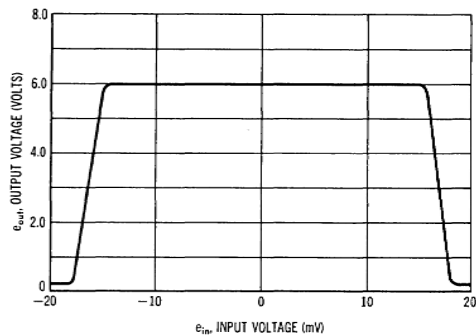
**FIGURE 7 — DIFFERENTIAL MODE  
RECOVERY TIME TEST CIRCUIT**



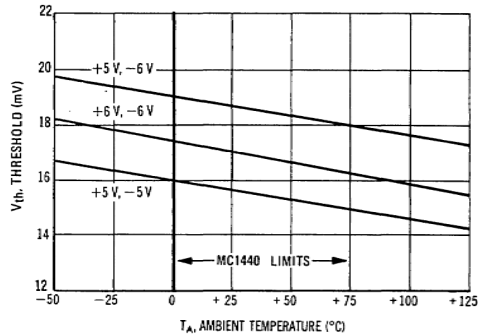
**FIGURE 8 — COMMON MODE  
RECOVERY TIME TEST CIRCUIT**



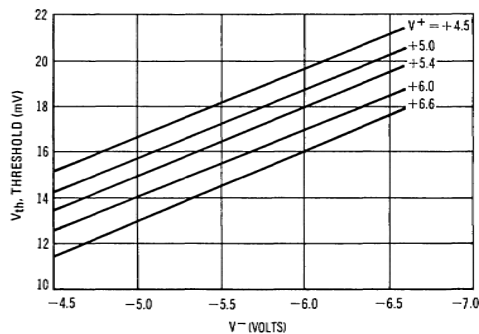
**FIGURE 9 — TYPICAL TRANSFER CHARACTERISTICS**



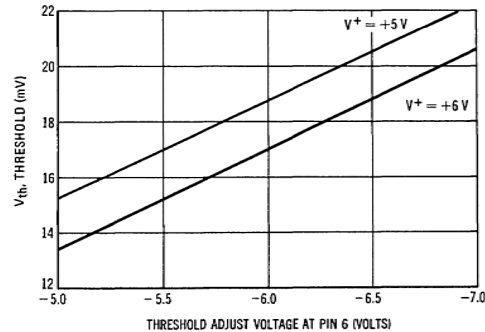
**FIGURE 10 — TYPICAL THRESHOLD versus TEMPERATURE**



**FIGURE 11 — TYPICAL THRESHOLD versus POWER SUPPLIES  
TA = +25°C (Threshold Adjust Attached to V<sup>-</sup>)**



**FIGURE 12 — TYPICAL THRESHOLD versus THRESHOLD  
VOLTAGE ADJUST FOR V<sup>-</sup> = 6.0V**



For a more detailed discussion regarding application of sense amplifiers, see Motorola Application Note AN-245, "The MC1540 — An Integrated Core Memory Sense Amplifier."

# MC1541 MC1441

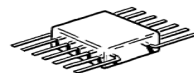
## SENSE AMPLIFIERS

Dual-channel gated sense amplifier with separate wideband differential input amplifiers. Either input can be gated on from saturated logic levels. The sense amplifier features adjustable threshold, saturated logic output levels, and a strobe input that accommodates saturated logic levels. Designed to detect bipolar signals from either of two sense lines. Operates with core memory cycle times less than 0.5  $\mu$ s.

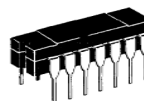
### Typical Amplifier Features:

- Nominal Threshold – 17 mV
- Input Offset Voltage – 1.0 mV typical
- Propagation Delay
  - Input to Gate-Output – 20 ns
  - Input to Amplifier-Output – 10 ns
  - Gate Response Time – 15 ns
  - Strobe Response Time – 15 ns
- Common Mode Input Range – 1.5 Volts
- Differential Mode Input Range
  - With Gate On – 600 mV
  - With Gate Off – 1.5 Volts
- Power Dissipation – 140 mW typical

See Packaging Information Section for outline dimensions.



**F SUFFIX**  
CERAMIC PACKAGE  
CASE 607  
TO-86

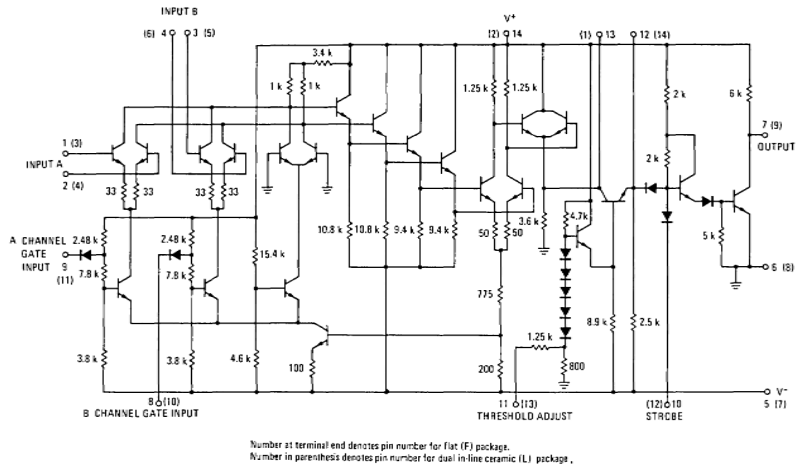


**L SUFFIX**  
CERAMIC PACKAGE  
CASE 632  
TO-116

### MAXIMUM RATINGS

| Rating                                 | Symbol         | Value                   | Unit  |
|----------------------------------------|----------------|-------------------------|-------|
| Power Supply Voltage                   | $V^+$<br>$V^-$ | +10<br>-10              | Vdc   |
| Differential Input Signal              | $V_{in}$       | $\pm 5$                 | Vdc   |
| Common Mode Input Voltage              | $CMV_{in}$     | $\pm 5$                 | Vdc   |
| Load Current                           | $I_L$          | 25                      | mA    |
| Power Dissipation (Package Limitation) | $P_D$          |                         |       |
| Flat Package                           |                | 500                     | mW    |
| Derate above 25°C                      |                | 3.3                     | mW/°C |
| Ceramic Dual In-Line Package           |                | 600                     | mW    |
| Derate above 25°C                      |                | 4.8                     | mW/°C |
| Operating Temperature Range            | $T_A$          | -55 to +125<br>0 to +75 | °C    |
| Storage Temperature Range              | $T_{stg}$      | -65 to +150             | °C    |

CIRCUIT SCHEMATIC



LOGIC DIAGRAM

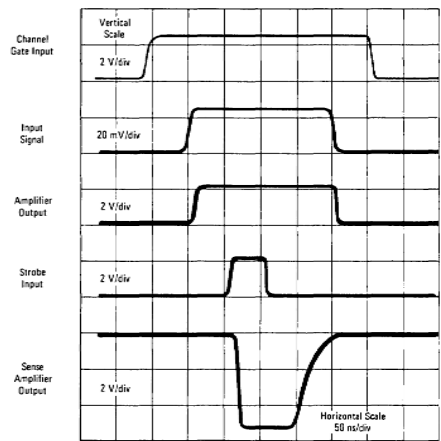
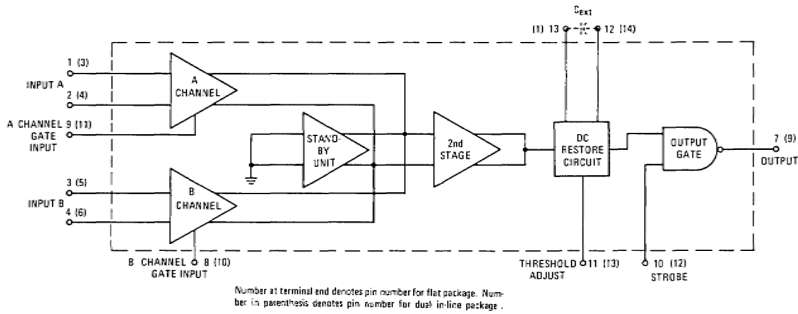


FIGURE 1 – TYPICAL OPERATION

## MC1541, MC1441 (continued)

### ELECTRICAL CHARACTERISTICS

( $V^+ = +5.0 \text{ Vdc} \pm 1\%$ ,  $V^- = 5.0 \text{ Vdc} \pm 1\%$ ,  $V_{th}(\text{pin } 11) = -5.0 \text{ Vdc} \pm 1\%$ ,  $C_{ext} = 0.01 \mu\text{F}$ ,  $T_A = 25^\circ\text{C}$  unless otherwise noted)  
 ( $T_{low} = -55^\circ\text{C}$  for MC1541 or  $0^\circ\text{C}$  for MC1441,  $T_{high} = +125^\circ\text{C}$  for MC1541 or  $+75^\circ\text{C}$  for MC1441. Pin numbers referenced in table denote flat package; to ascertain corresponding pin number for dual in-line package refer to the equivalent circuit.)

| Characteristic                                                                                                                                                                                 | Fig. No. | Symbol               | Min            | Typ                    | Max            | Unit            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------|----------------|------------------------|----------------|-----------------|
| Input Threshold Voltage<br>( $T_A = +25^\circ\text{C}$ )<br>( $T_{low} \leq T_A \leq T_{high}$ )                                                                                               | 8        | $V_{th}$             | 14<br>13<br>12 | 17<br>-<br>17          | 20<br>21<br>22 | mV              |
| Input Offset Voltage                                                                                                                                                                           | 8        | $V_{io}$             | -              | 1.0                    | 6.0            | mV              |
| Input Bias Current<br>( $V_1 = V_2 = V_3 = V_4 = 0$ )<br>( $V_1 = V_2 = V_3 = V_4 = 0$ , $T_A = T_{low}$ )                                                                                     | 9        | $I_b$                | -<br>-         | 5.0<br>-               | 25<br>50       | $\mu\text{A}$   |
| Input Offset Current                                                                                                                                                                           | 9        | $I_{io}$             | -              | 1.0                    | 2.0            | $\mu\text{A}$   |
| Output Voltage High<br>( $V_1 = V_2 = V_3 = V_4 = 0$ , $I_{OH} = 200 \mu\text{A}$ )                                                                                                            |          | $V_{OH}$             | 3.0            | -                      | -              | Vdc             |
| Output Voltage Low<br>( $V_1 = V_2 = V_3 = V_4 = 0$ , $V_{12} = +5.0 \text{ Vdc}$ , $I_7 = 10 \text{ mAdc}$ )<br>( $V_{12} = +5.0 \text{ Vdc}$ , $I_7 = 10 \text{ mAdc}$ , $T_A = +T_{high}$ ) | 10       | $V_{OL}$             | -<br>-         | -<br>-                 | 350<br>400     | mVdc            |
| Strobe Load Current<br>( $V_{10} = 0$ )                                                                                                                                                        |          | $I_S$                | -              | -                      | 1.5            | mAdc            |
| Strobe Reverse Current<br>( $V_{10} = +5.0 \text{ Vdc}$ )<br>( $V_{10} = +5.0 \text{ Vdc}$ , $T_A = T_{high}$ )                                                                                |          | $I_{SR}$             | -<br>-         | -<br>-                 | 2.0<br>25      | $\mu\text{Adc}$ |
| Input Gate Voltage Low<br>( $V_1 = V_3 = 25 \text{ mVdc}$ , $V_2 = V_4 = 0$ )                                                                                                                  | 11       | $V_{GL}$             | -              | 0.7                    | -              | Vdc             |
| Input Gate Voltage High<br>( $V_1 = V_3 = 25 \text{ mVdc}$ , $V_2 = V_4 = 0$ )                                                                                                                 | 11       | $V_{GH}$             | -              | 1.6                    | -              | Vdc             |
| Input Gate Load Current<br>( $V_8$ or $V_9 = 0$ )                                                                                                                                              |          | $I_G$                | -              | -                      | 2.5            | mAdc            |
| Input Gate Reverse Current ( $V_8$ or $V_9 = 5.0 \text{ Vdc}$ )<br>( $T_A = 25^\circ\text{C}$ )<br>( $T_A = T_{high}$ )                                                                        |          | $I_{GR}$             | -<br>-         | -<br>-                 | 2.0<br>25      | $\mu\text{Adc}$ |
| Common Mode Range<br>Input Gate High<br>Input Gate Low                                                                                                                                         | 13       | $V_{CM}$             | -<br>-         | $\pm 1.5$<br>$\pm 1.5$ | -<br>-         | Vdc             |
| Differential Mode Range<br>Input Gate High<br>Input Gate Low                                                                                                                                   | 14       | $V_{DH}$<br>$V_{DL}$ | -<br>-         | $\pm 600$<br>$\pm 1.5$ | -<br>-         | mV<br>Vdc       |
| Power Dissipation                                                                                                                                                                              |          | $P_D$                | -              | 140                    | 180            | mW              |

### SWITCHING CHARACTERISTICS

| Characteristic                                                                                                    | Fig. No. | Symbol    | Min    | Typ      | Max      | Unit |
|-------------------------------------------------------------------------------------------------------------------|----------|-----------|--------|----------|----------|------|
| Propagation Delay<br>Input to Amplifier Output<br>( $V_1 = 25 \text{ mV}$ pulse, $V_{10} = +2.0 \text{ Vdc}$ )    | 8        | $t_{IA}$  | -      | 10       | 15       | ns   |
| Input to Output<br>( $V_1 = 25 \text{ mV}$ pulse, $V_{10} = +2.0 \text{ Vdc}$ )                                   | 8        | $t_{IO}$  | -      | 20       | 30       |      |
| Strobe to Output<br>( $V_1 = V_2 = V_3 = V_4 = 0$ , $V_{10} = +2.0 \text{ V}$ pulse)                              | 12       | $t_{SO}$  | -      | 15       | 20       |      |
| Gate Input to Amplifier Input<br>( $V_1 = 25 \text{ mV}$ pulse, $V_9 = 2.0 \text{ V}$ pulse)                      | 11       | $t_{GI}$  | -      | 10       | 15       |      |
| Gate Input to Amplifier Output<br>( $V_1 = 25 \text{ mVdc}$ , $V_9 = 2.0 \text{ V}$ pulse)                        | 11       | $t_{GA}$  | -      | 30       | 35       |      |
| Recovery Time<br>Differential Mode<br>Input Gate High } $V_1$ or $V_3 = 400 \text{ mV}$ pulse<br>Input Gate Low } | 14       | $t_{DR}$  | -<br>- | 30<br>0  | -<br>-   | ns   |
| Common Mode<br>Input Gate High } $V_1$ or $V_3 = 1.5 \text{ V}$ pulse<br>Input Gate Low }                         | 13       | $t_{CMR}$ | -<br>- | 15<br>15 | 30<br>30 |      |

FIGURE 2 – TYPICAL INPUT THRESHOLD versus TEMPERATURE

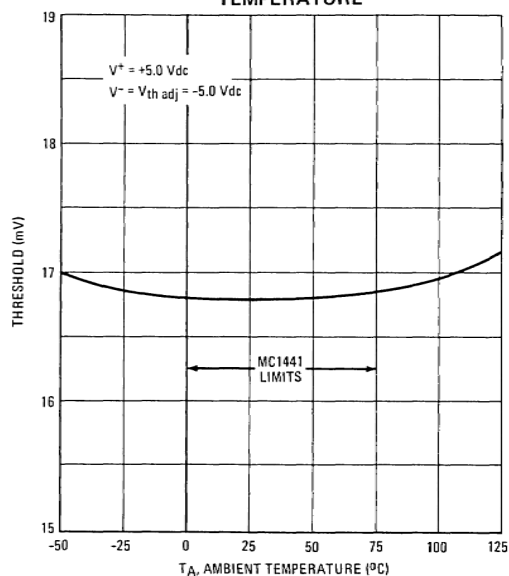


FIGURE 3 – TYPICAL THRESHOLD versus THRESHOLD VOLTAGE ADJUST

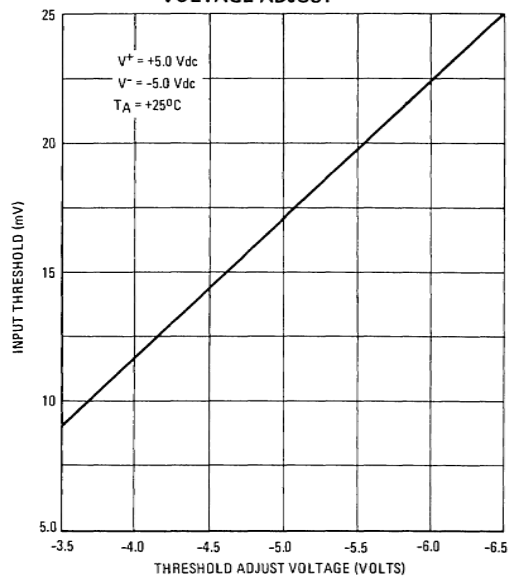


FIGURE 4 – TYPICAL INPUT THRESHOLD versus  $V^-$

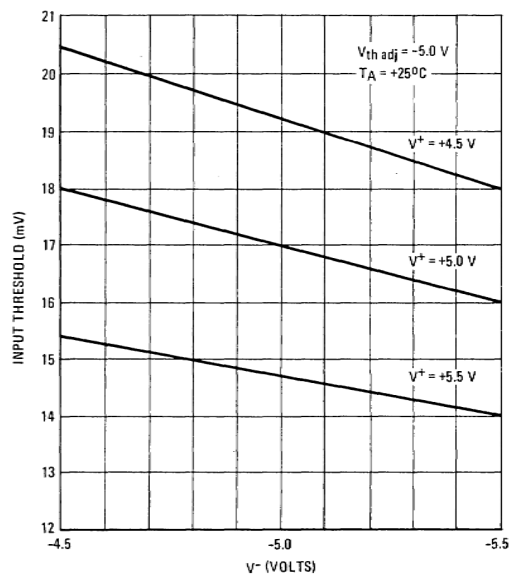


FIGURE 5 – TYPICAL INPUT THRESHOLD versus INPUT PULSE WIDTH

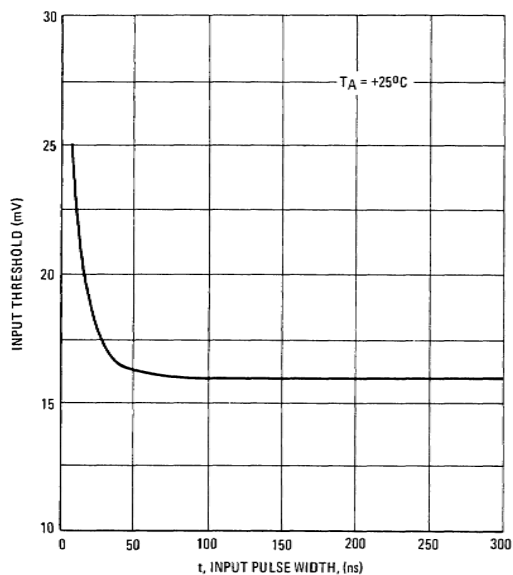




FIGURE 6 – INPUT-OUTPUT TRANSFER CHARACTERISTICS

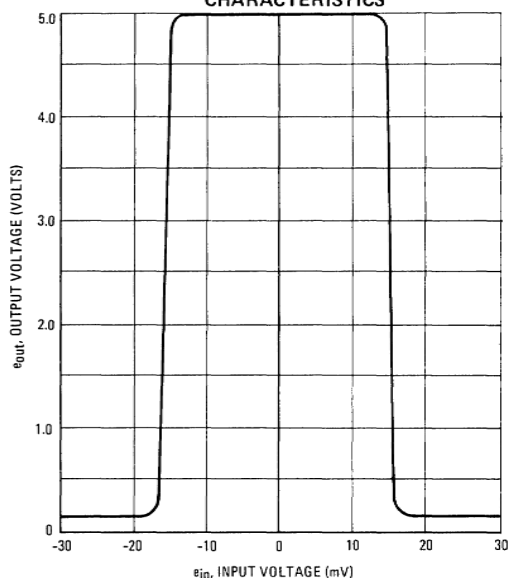


FIGURE 7 – CHANNEL GATE INPUT-AMPLIFIER OUTPUT TRANSFER CHARACTERISTICS

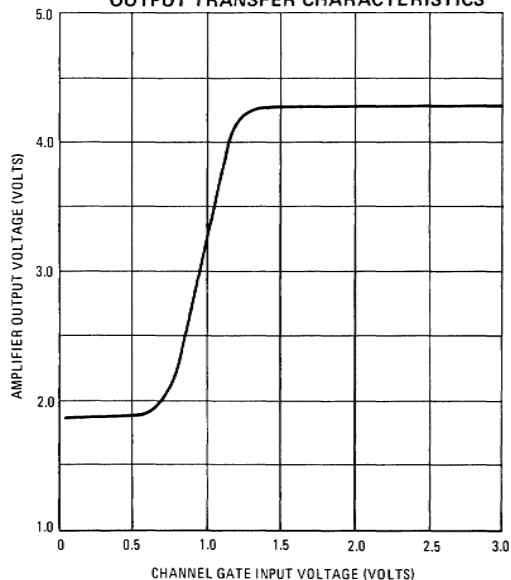
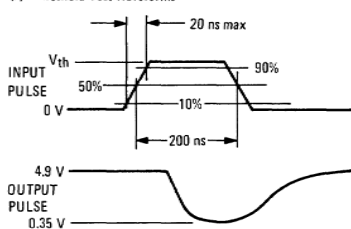
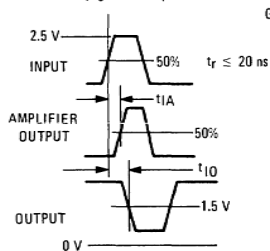


FIGURE 8 – INPUT THRESHOLD FOR OUTPUT VOLTAGE SWING FROM  $V_{OH}$  TO  $V_{OL}$  PROPAGATION DELAY FROM INPUT TO OUTPUT

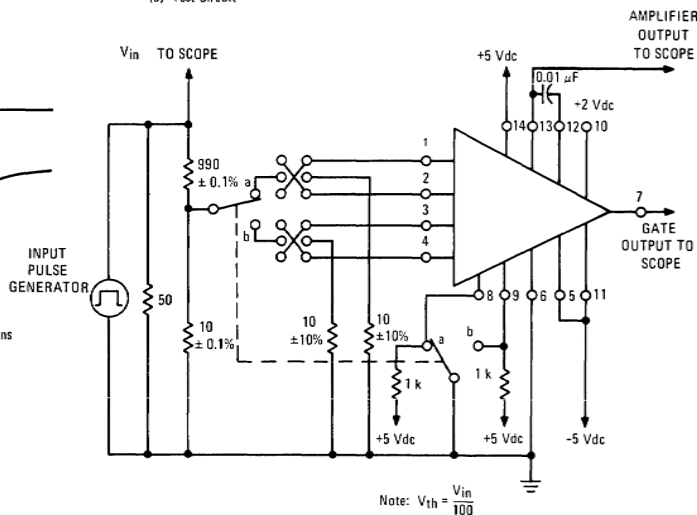
(a) Threshold Test Waveforms



(c) Waveforms for Propagation Delay Test



(b) Test Circuit



Number at terminal end denotes the pin number for flat package only; to ascertain the corresponding pin number for the dual in line packages refer to the circuit schematic on the second page.

FIGURE 9 – INPUT BIAS CURRENT TEST CIRCUIT

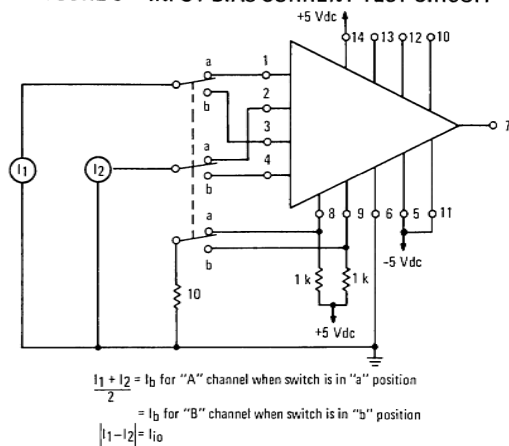


FIGURE 10 – OUTPUT VOLTAGE LEVELS

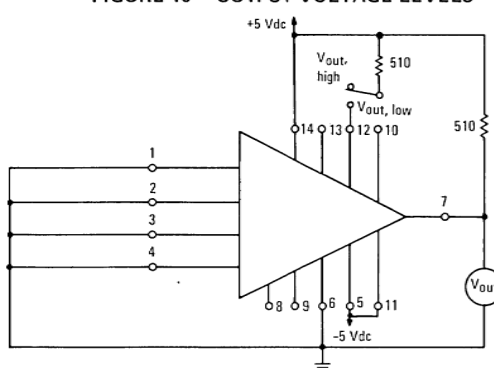
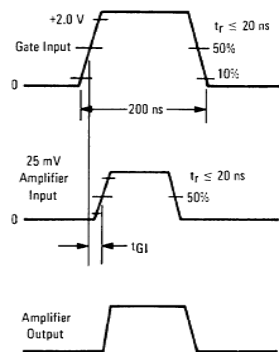
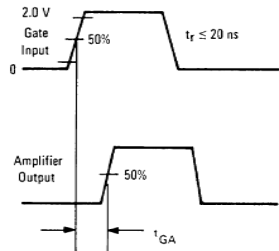


FIGURE 11 – MINIMUM TIME FROM CHANNEL GATE INPUT TO AMPLIFIER INPUT  
 PROPAGATION DELAY FROM CHANNEL GATE INPUT TO AMPLIFIER OUTPUT

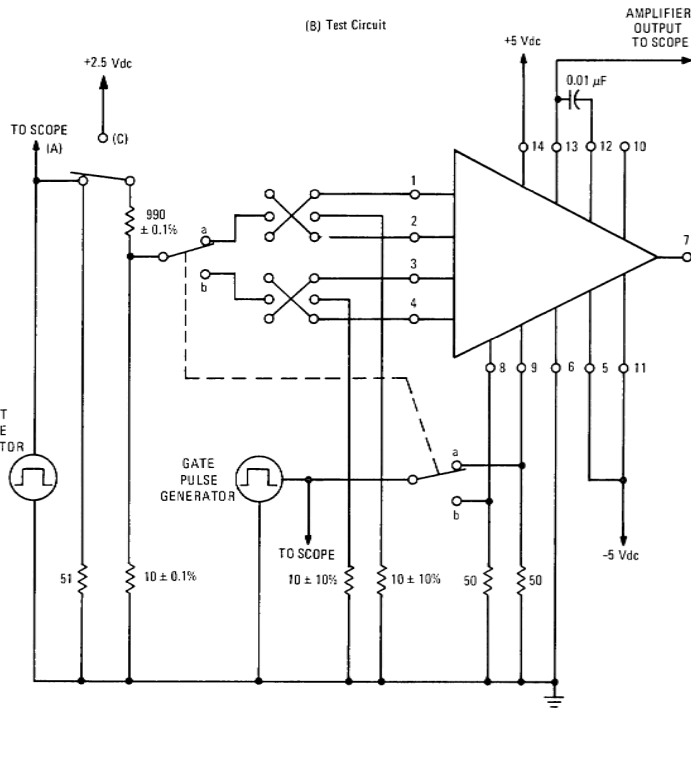
(A) Minimum Time from Gate Input to Amplifier Input –  $t_{GI}$   
 (See Definitions)



(C) Propagation Delay from Channel Gate Input to Amplifier Output



(B) Test Circuit



(Pin numbers shown on this page denote the pin numbers for the flat package only; to ascertain the corresponding pin numbers for the dual in-line package, refer to the circuit schematic on the second page.)

FIGURE 12 – PROPAGATION DELAY FROM STROBE INPUT TO OUTPUT

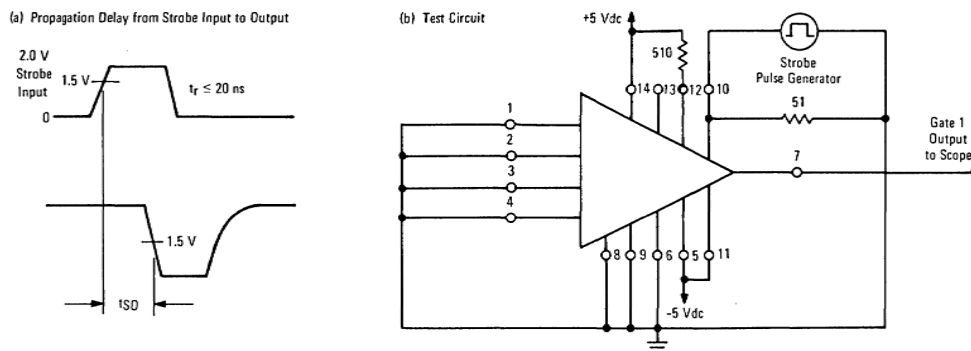


FIGURE 13 – COMMON-MODE RECOVERY AND COMMON-MODE RANGE

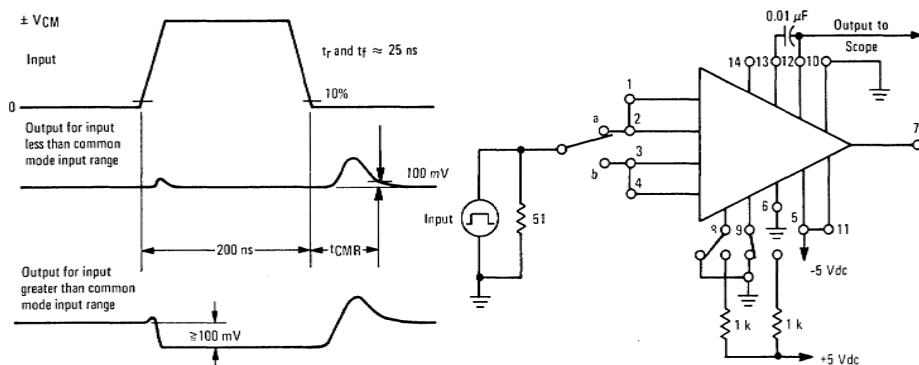
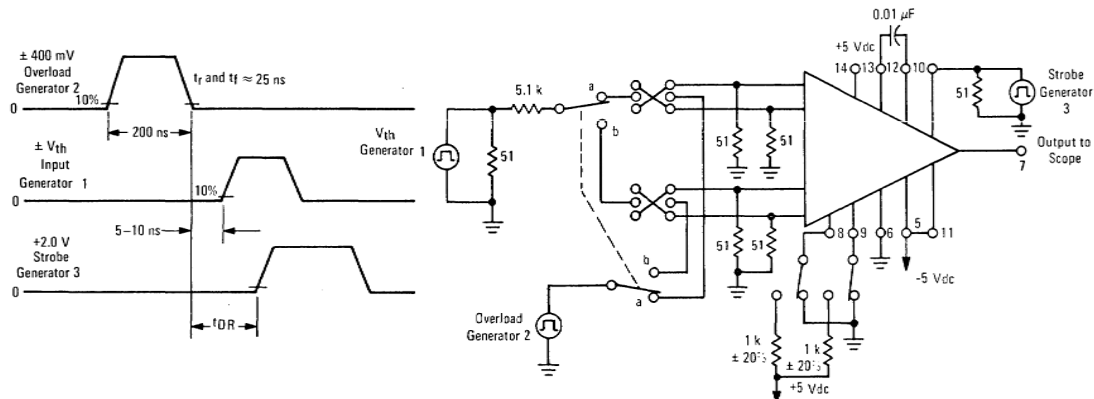


FIGURE 14 – DIFFERENTIAL RECOVERY AND DIFFERENTIAL RANGE



(Pin numbers shown on this page denote the pin numbers for the flat package only; to ascertain the corresponding pin numbers for the dual in-line package, refer to the circuit schematic on the second page.)

## DEFINITIONS

Pin numbers referenced in the definitions below denote the flat package only; to ascertain the corresponding pin number for the dual in-line package refer to the circuit schematic.

|                        |                                                                                                                                                                                                                                                                                                                                                               |                       |                                                                                                                                                                                                      |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>I<sub>B</sub></b>   | Input Bias Current — The average input current defined as $(I_1 + I_2 + I_3 + I_4)/4$ .                                                                                                                                                                                                                                                                       |                       |                                                                                                                                                                                                      |
| <b>I<sub>G</sub></b>   | Channel Gate Load Current — The amount of current drain from the circuit when the channel gate input (Pin 8 or 9) is grounded.                                                                                                                                                                                                                                | <b>t<sub>IO</sub></b> | Propagation Delay, Input to Output — The time required for the gate output pulse at pin 7 to reach the 1.5 Volt level as referenced to 50% of the input pulse at pins 1 and 2 or 3 or 4.             |
| <b>I<sub>GR</sub></b>  | Channel Gate Reverse Current — The leakage current when the channel gate input (Pin 8 or 9) is high.                                                                                                                                                                                                                                                          | <b>t<sub>SO</sub></b> | Strobe Propagation Delay to Output — The time required for the output pulse at pin 7 to reach the 1.5 Volt level as referenced to the 1.5 Volt level of the strobe input at pin 10.                  |
| <b>I<sub>IO</sub></b>  | Input Offset Current — The difference between amplifier input current values $ I_1 - I_2 $ or $ I_3 - I_4 $ .                                                                                                                                                                                                                                                 | <b>V<sub>CM</sub></b> | Maximum Common Mode Input Range — The common mode input voltage which causes the output voltage level of the amplifier to decrease by 100 mV. (This is independent of the channel gate input level.) |
| <b>I<sub>S</sub></b>   | Strobe Load Current — The amount of current drain from the circuit when the strobe pin is grounded.                                                                                                                                                                                                                                                           | <b>V<sub>DH</sub></b> | Maximum Differential Input Range, Gate Input High — The differential input which causes the input stage to begin saturation.                                                                         |
| <b>I<sub>SR</sub></b>  | Strobe Reverse Current — The leakage current when the strobe input is high.                                                                                                                                                                                                                                                                                   | <b>V<sub>DL</sub></b> | Maximum Differential Input Range, Gate Input Low — The differential input signal which causes the output voltage level of the amplifier to decrease by 100 mV.                                       |
| <b>P<sub>D</sub></b>   | Power Dissipation — The amount of power dissipated in the unit.                                                                                                                                                                                                                                                                                               | <b>V<sub>GH</sub></b> | Channel Gate Input Voltage High — Gate pulse amplitude that allows the amplifier output pulse to just reach 100% of its final value. (Amplifier input is set at 25 mVdc).                            |
| <b>t<sub>CMR</sub></b> | Common Mode Recovery Time — The time required for the voltage at pin 12 to be within 100 mV of the dc value (after overshoot or ringing) as referenced to the 10% point of the trailing edge of a common mode overload signal.                                                                                                                                | <b>V<sub>GL</sub></b> | Channel Gate Input Voltage Low — Gate pulse amplitude that allows the amplifier output to just reach a 100 mV level. (Amplifier input is set at 25 mVdc).                                            |
| <b>t<sub>DR</sub></b>  | Differential Recovery Time — The time required for the device to recover from the specified differential input prior to strobe enable as referenced to the 10% point of the trailing edge of an input pulse. The device is considered recovered when the threshold with the overload signal applied is within 1.0 mV of the threshold with no overload input. | <b>V<sub>IO</sub></b> | Input Offset Voltage — The difference in $V_{th}$ between inputs at pins 1 and 2 or 3 and 4.                                                                                                         |
| <b>t<sub>GI</sub></b>  | Minimum Time Between Channel Gate Input and Signal Input — The minimum time between 50% point of channel gate input (Pin 8 or 9) and 50% point of signal input (Pins 1, 2, 3, or 4) that still allows a full width signal at amplifier output.                                                                                                                | <b>V<sub>OH</sub></b> | Output Voltage High — The high-level output voltage when the output gate is turned off.                                                                                                              |
| <b>t<sub>GA</sub></b>  | Propagation Delay, Channel Gate Input to Amplifier Output — The time required for the amplifier output at pin 13 to reach 50% of its final value as referenced to 50% of the input gate pulse at pin 8 or 9 (Amplifier input = 25 mVdc).                                                                                                                      | <b>V<sub>OL</sub></b> | Output Voltage Low — The low-level output voltage when the output gate is saturated and the output sink current is 10 mA.                                                                            |
| <b>t<sub>IA</sub></b>  | Propagation Delay, Input to Amplifier Output — The time required for the amplifier output pulse at pin 13 to achieve 50% of its final value referenced to 50% of the input pulse at pins 1 and 2 or 3 and 4.                                                                                                                                                  | <b>V<sub>th</sub></b> | Input Threshold — Input pulse amplitude at pins 1, 2, 3 or 4 that causes the output gate to just reach $V_{OL}$ .                                                                                    |

# MC1543L

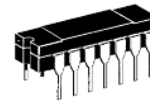
## DUAL SENSE AMPLIFIER

### DUAL MECL CORE-MEMORY SENSE AMPLIFIER

A dual dc coupled sense amplifier. Output levels are compatible with emitter coupled logic levels. MC1543L offers adjustable threshold and excellent threshold stability over a wide range of power-supply voltage variation.

#### Typical Amplifier Features:

- Input Threshold  
Adjustable from 10 to 40 mV (Positive or Negative Signals)
- Both OR and NOR Outputs Available
- Low Power Dissipation
- Threshold Insensitive to + or - Supply Variation
- Each Amplifier is Separately Strobed

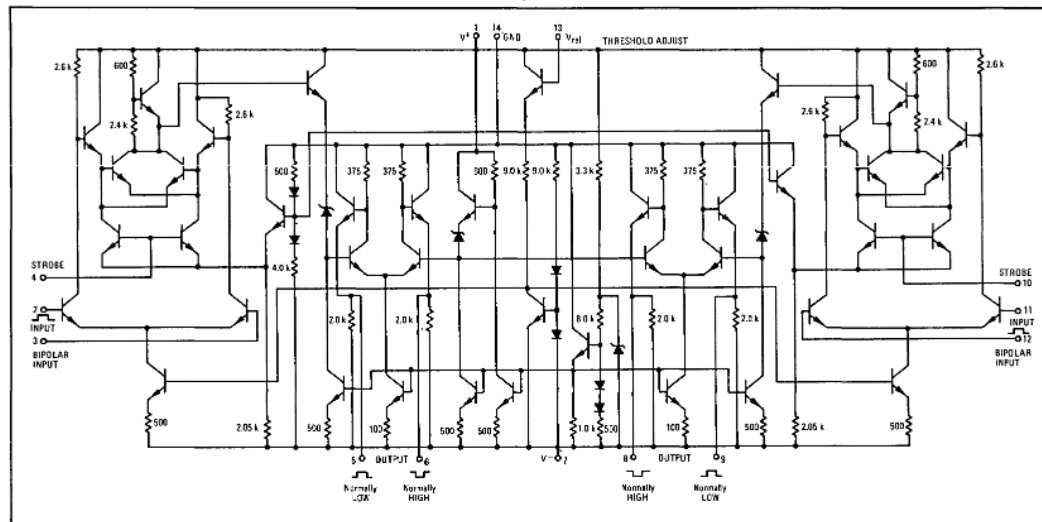


CERAMIC PACKAGE  
CASE 632  
TO-116

#### MAXIMUM RATINGS ( $T_A = 25^\circ\text{C}$ unless otherwise noted)

| Rating                                 | Symbol         | Value        | Unit                       |
|----------------------------------------|----------------|--------------|----------------------------|
| Power Supply Voltage                   | $V^+$<br>$V^-$ | +10<br>-10   | Vdc<br>Vdc                 |
| Differential Input Signal              | $V_{in}$       | $\pm 5.0$    | Vdc                        |
| Common Mode Input Voltage              | $CMV_{in}$     | $\pm 5.0$    | Vdc                        |
| Load Current                           | $I_L$          | 25           | mA                         |
| Power Dissipation (Package Limitation) | $P_D$          | 100/0<br>6.7 | mW<br>mW/ $^\circ\text{C}$ |
| Operating Temperature Range            | $T_A$          | -55 to +125  | $^\circ\text{C}$           |
| Storage Temperature Range              | $T_{stg}$      | -65 to +150  | $^\circ\text{C}$           |

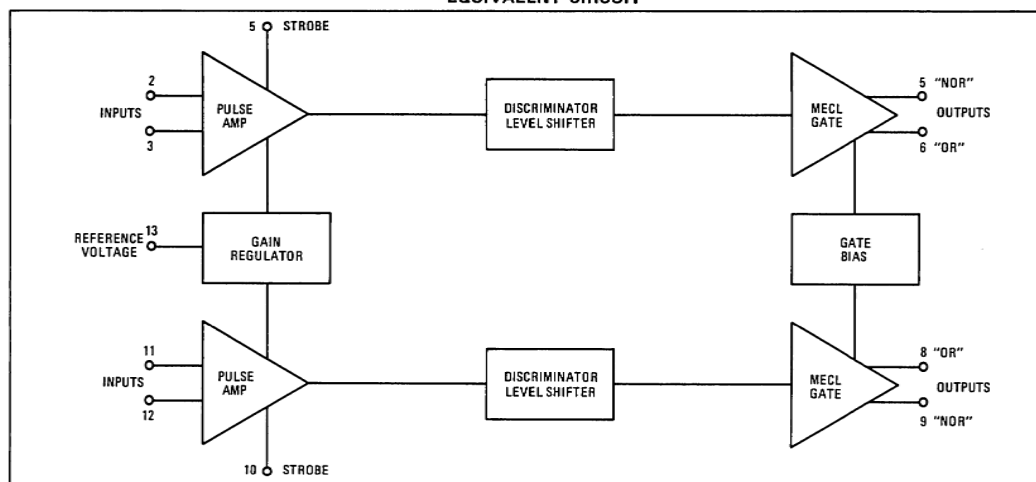
#### CIRCUIT SCHEMATIC



See Packaging Information Section for outline dimensions.

# MC1543L (continued)

EQUIVALENT CIRCUIT



## ELECTRICAL CHARACTERISTICS (Each Amplifier)

( $V^+ = +5.0 \text{ Vdc} \pm 5\%$ ,  $V^- = -5.2 \text{ Vdc} \pm 5\%$ ,  $V_{\text{ref}} = 0.54 \text{ V} \pm 1\%$ ,  $T_A = +25^\circ\text{C}$  unless otherwise noted)

| Characteristic                                                          | Fig. No. | Symbol           | Min   | Typ   | Max   | Unit            |
|-------------------------------------------------------------------------|----------|------------------|-------|-------|-------|-----------------|
| Input Threshold Voltage                                                 | 8        | $V_{\text{th}}$  | 17    | 20    | 23    | mV              |
| Power Supply Currents<br>( $V_2 = V_3 = V_{11} = V_{12} = V_{14} = 0$ ) | 6        | $I_{\text{CC}}$  | —     | 9.5   | 12    | mAdc            |
|                                                                         | 6        | $I_{\text{EE}}$  | —     | 26.5  | 33    | mAdc            |
| Input Bias Current                                                      | 7        | $I_b$            | —     | 3.5   | 10    | $\mu\text{Adc}$ |
| Input Offset Current                                                    | 7        | $I_{\text{io}}$  | —     | 0.05  | 0.5   | $\mu\text{Adc}$ |
| Output Voltage High                                                     | 9        | $V_{\text{OH}}$  | -0.85 | -0.8  | -0.67 | Vdc             |
| Output Voltage Low                                                      | 9        | $V_{\text{OL}}$  | —     | -1.7  | -1.46 | Vdc             |
| Strobe Threshold Level                                                  | 10       | $V_{\text{ST}}$  | —     | -1.30 | —     | Vdc             |
| Strobe Input Current High                                               | 10       | $I_{\text{SH}}$  | —     | 25    | 50    | $\mu\text{Adc}$ |
| Strobe Input Current Low                                                | 10       | $I_{\text{SL}}$  | —     | 0.01  | 0.1   | $\mu\text{Adc}$ |
| Input Common Mode Range                                                 | 14       | $V_{\text{CM}}$  | 3.0   | 4.0   | —     | Vdc             |
| Input Threshold Range (by varying $V_{\text{ref}}$ )                    | 8        | $V_{\text{thR}}$ | —     | 10-40 | —     | mV              |
| Power Dissipation                                                       | 6        | $P_D$            | —     | 185   | 230   | mW              |
| Reference Supply Input Current (Pin 13)                                 | 6        | $I_{\text{ref}}$ | —     | 10    | 40    | $\mu\text{A}$   |

## SWITCHING CHARACTERISTICS

|                                                                             |    |                  |   |     |    |    |
|-----------------------------------------------------------------------------|----|------------------|---|-----|----|----|
| Propagation Delay (Input to Output)                                         | 11 | $t_{\text{PO}}$  | — | 28  | 35 | ns |
| Propagation Delay (Strobe to Output)                                        | 12 | $t_{\text{SO}}$  | — | 16  | 20 | ns |
| Strobe Release Time                                                         | 12 | $t_{\text{SR}}$  | — | 18  | 30 | ns |
| Recovery Time (Differential Mode)<br>( $e_{\text{in}} = 400 \text{ mVdc}$ ) | 13 | $t_{\text{DR}}$  | — | 10  | 15 | ns |
| Recovery Time (Common Mode)<br>( $e_{\text{in}} = 4.0 \text{ Vdc}$ )        | 14 | $t_{\text{CMR}}$ | — | 3.0 | 15 | ns |
| Strobe Width Minimum                                                        | 12 | $t_{\text{S}}$   | — | 8.0 | —  | ns |

## TEMPERATURE TESTS ( $-55^\circ\text{C}$ to $+125^\circ\text{C}$ )

|                         |                                                                         |   |                 |          |              |          |                 |
|-------------------------|-------------------------------------------------------------------------|---|-----------------|----------|--------------|----------|-----------------|
| Input Threshold Voltage | $\begin{cases} (-55^\circ\text{C}) \\ (+125^\circ\text{C}) \end{cases}$ | 8 | $V_{\text{th}}$ | 18<br>15 | 21.5<br>18.5 | 25<br>22 | mV              |
| Input Bias Current      |                                                                         | 7 | $I_b$           | 2.2      | 7.0          | 20       | $\mu\text{Adc}$ |
| Input Offset Current    |                                                                         | 7 | $I_{\text{io}}$ | 0.02     | 0.1          | 1.0      | $\mu\text{Adc}$ |

TYPICAL CHARACTERISTICS

FIGURE 1 – TYPICAL INPUT THRESHOLD  
versus TEMPERATURE

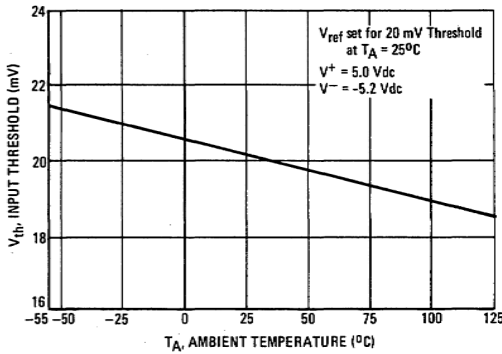


FIGURE 2 – TYPICAL INPUT THRESHOLD  
versus REFERENCE VOLTAGE

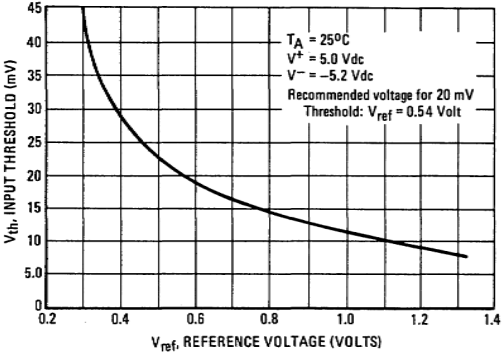


FIGURE 3A – TYPICAL INPUT THRESHOLD versus  $V^+$

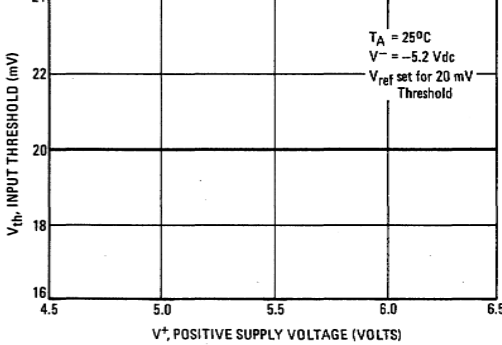


FIGURE 3B – TYPICAL INPUT THRESHOLD versus  $V^-$

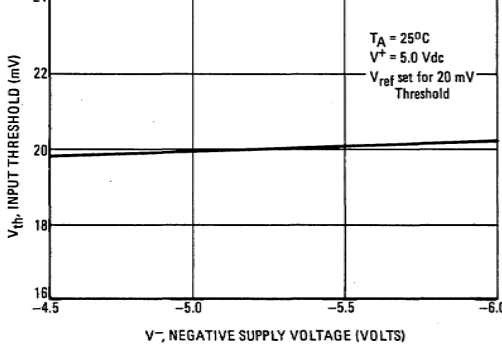


FIGURE 4 – TYPICAL INPUT THRESHOLD  
versus INPUT PULSE WIDTH

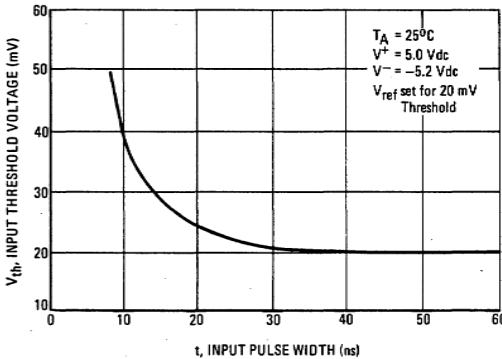
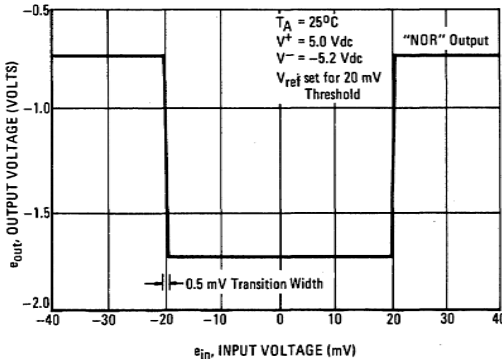


FIGURE 5 – INPUT-OUTPUT TRANSFER CHARACTERISTICS  
(one output)



# MC1543L (continued)

FIGURE 6 – POWER SUPPLY CURRENT DRAIN

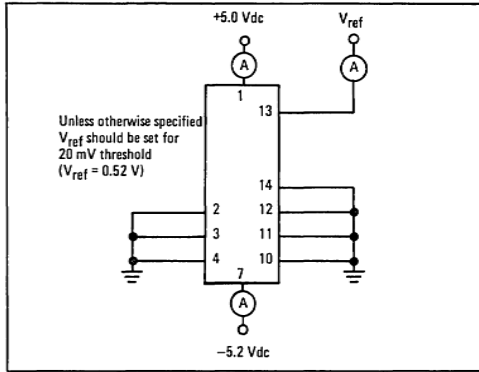


FIGURE 7 – INPUT BIAS CURRENT  
INPUT OFFSET CURRENT

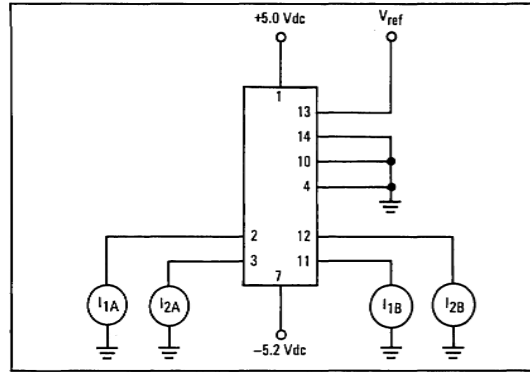


FIGURE 8 – INPUT THRESHOLD LEVEL

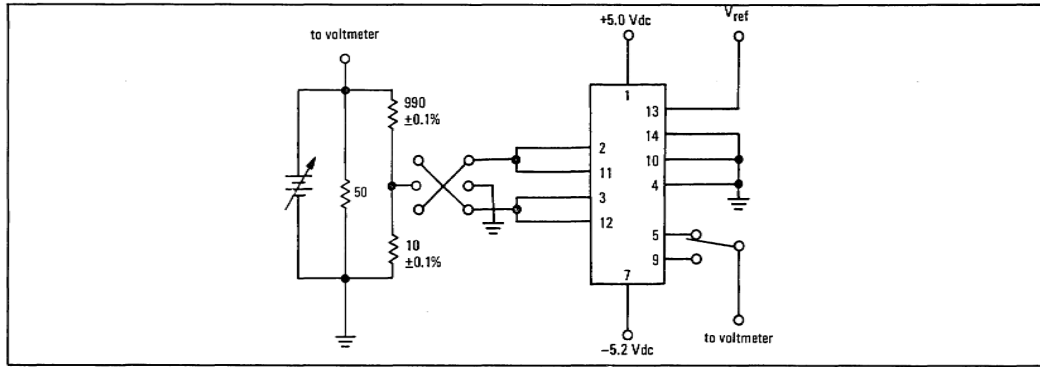


FIGURE 9 – OUTPUT VOLTAGE LEVELS

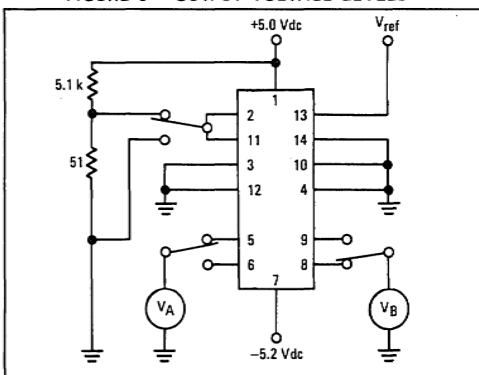


FIGURE 10 – STROBE THRESHOLD LEVEL  
STROBE INPUT CURRENTS

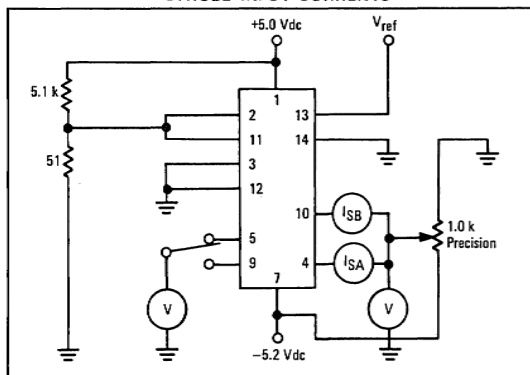




FIGURE 11 — PROPAGATION DELAY —  
INPUT TO OUTPUT

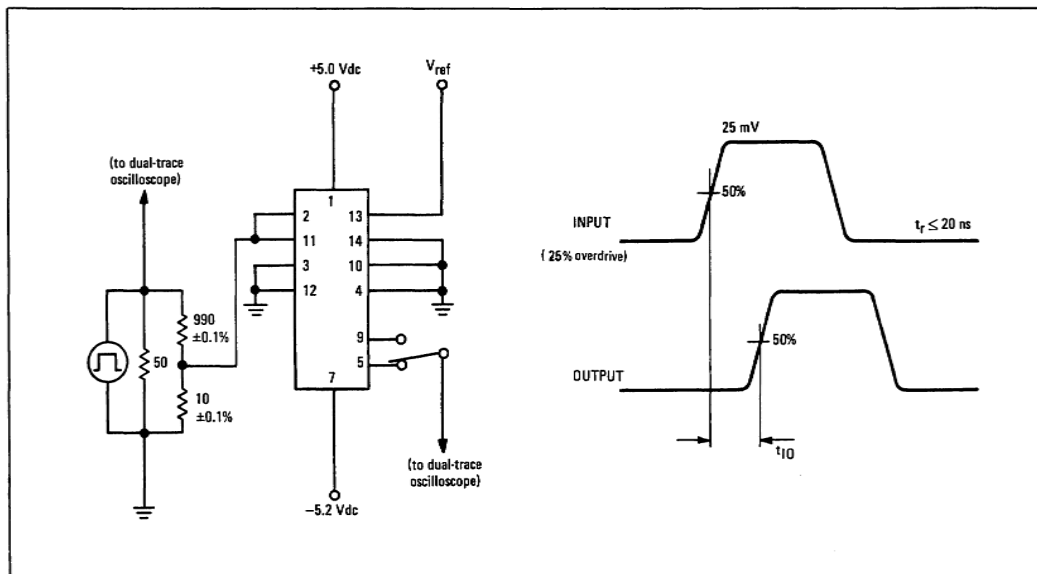


FIGURE 12 — PROPAGATION DELAY —  
STROBE TO OUTPUT and STROBE RELEASE TIME

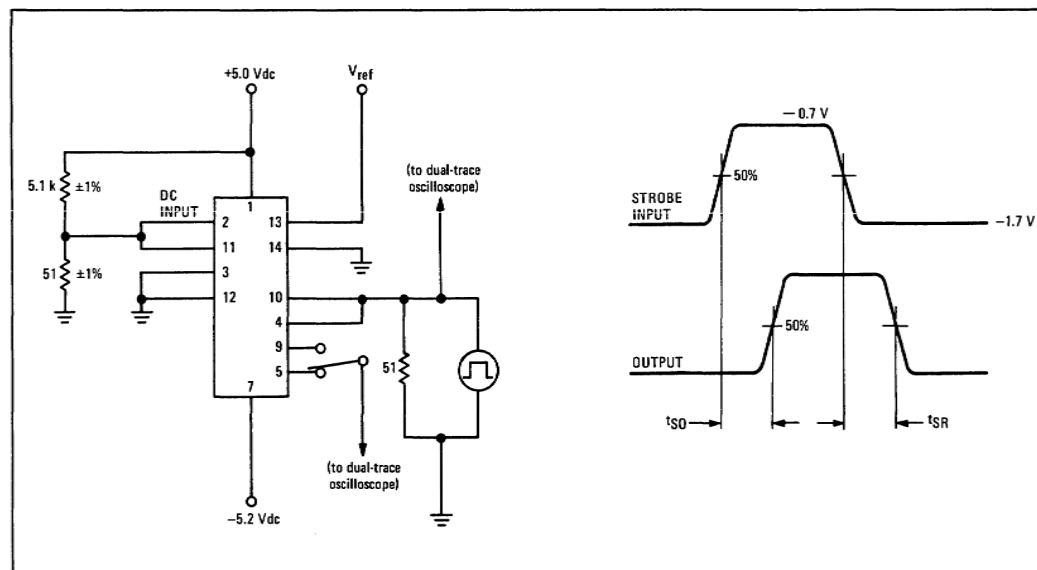


FIGURE 13 — DIFFERENTIAL MODE RECOVERY TIME  
(See definition section)

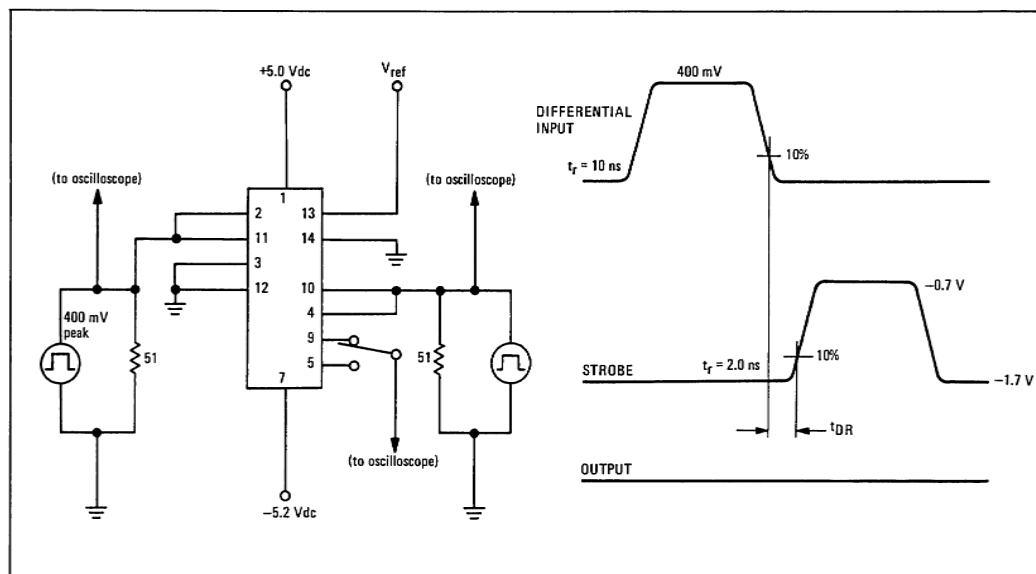
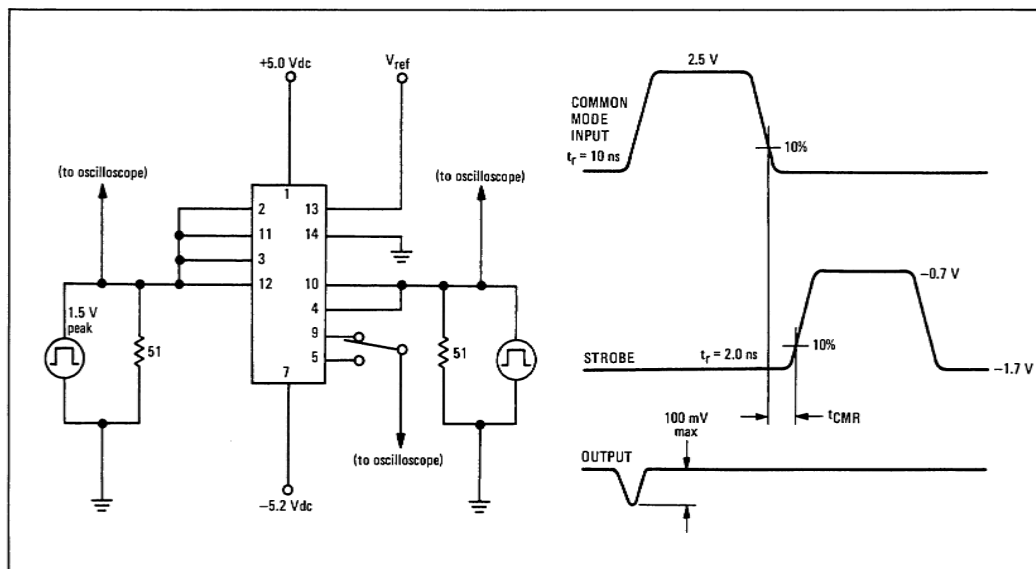


FIGURE 14 — COMMON MODE RECOVERY TIME  
COMMON MODE INPUT RANGE  
(See definition section)



## DEFINITIONS

- $I_{IO}$  Input Offset Current — The difference between amplifier input current values  $|I_{1A} - I_{2A}|$  or  $|I_{1B} - I_{2B}|$ .
- $I_{SH}$  Strobe High Current — The amount of input current when the strobe pin is grounded.
- $I_{SL}$  Strobe Low Current — The leakage current when the strobe input is tied to the negative supply.
- $P_D$  Power Dissipation — The amount of power dissipated in the unit.
- $t_{CMR}$  Common Mode Recovery Time — The minimum time by which the strobe input may follow the high level common mode input signal without causing a signal to appear at the amplifier output.
- $t_{DR}$  Differential Mode Recovery Time — Differential recovery time, the minimum time by which the strobe input may follow the high level differential input signal without causing a signal to appear at the amplifier output.
- $t_{IO}$  Propagation Delay, Amplifier Input to Amplifier Output — The time required for the amplifier output to reach 50% of its final value as referenced to 50% of the level of the pulse input (Amplifier input = 25 mVdc or 25% over set threshold).
- $t_S$  Strobe Width — The amount of time the strobe must be high to obtain a given output. Minimum strobe width is that minimum time required to cause the output to complete a full swing  $V_{OL}$  to  $V_{OH}$  or  $V_{OH}$  to  $V_{OL}$ .
- $t_{SO}$  Propagation Delay, Strobe Input to Amplifier Output — The time required for the amplifier output pulse to achieve 50% of its final value referenced to 50% of the strobe input pulse at pins 4 or 10.
- $t_{SR}$  Strobe Release Time — The time required for the output to change to 50% of its swing after the strobe reaches 50% of its level going low. A dc level of 50 mV is the input signal.
- $V_{CM}$  Maximum Common Mode Input Range — The common mode input voltage which causes the output voltage level of the amplifier to change by 100 mV (strobe high).
- $V_{OH}$  Output Voltage High — The high-level output voltage at pins 6 and 8 with no input — or at pins 5 and 9 with input above threshold.
- $V_{OL}$  Output Voltage Low — The low-level output voltage at pins 5 and 9 with no input — or at pins 6 and 8 with input above threshold.
- $V_{ST}$  Strobe Threshold Level — The voltage at which the strobe turns the amplifier to the ON state.
- $V_{th}$  Input Threshold — Input pulse amplitude at pins 2, 3, 11, or 12 that causes the output gate to just reach its new value,  $V_{OL}$  or  $V_{OH}$ .
- $V_{thR}$  Input Threshold Range — The maximum spread of input threshold level that can be attained by varying the threshold voltage reference,  $V_{ref}$ .

# MC1544L MC1444L

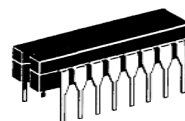
## SENSE AMPLIFIERS

### IDEAL FOR PLATED-WIRE, THIN-FILM AND OTHER HIGH-SPEED LOW-LEVEL SENSING APPLICATIONS

MC1544L/MC1444L features four input channels with decoded selection, two stages of gain employing capacitive coupling, and a MTTL compatible output gate. AC coupling reduces access times by eliminating the problems usually associated with input line offset voltages.

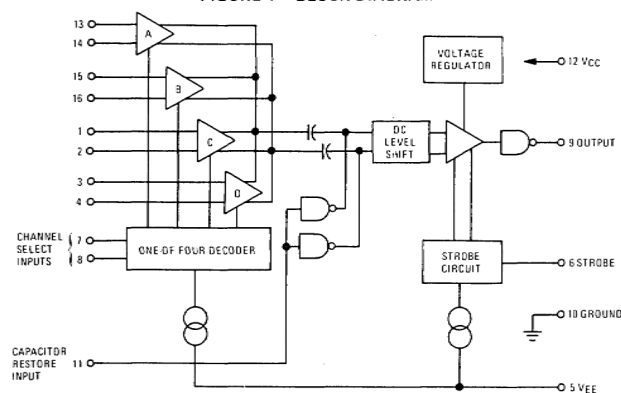
- Threshold Level — 1.0 mV typ
- Propagation Delay Time — 18 ns typ
- Decoded Input Channel Selection
- MTTL Compatible Inputs and Outputs
- Wired OR Output Capability
- DC Level Restore Gate on Capacitors Eliminates Repetition Rate Problems Common to ac-Coupled Circuits
- Output Strobe Capability

AC-COUPLED  
FOUR-CHANNEL  
SENSE AMPLIFIER  
MONOLITHIC SILICON  
EPITAXIAL PASSIVATED  
INTEGRATED CIRCUIT



CERAMIC PACKAGE  
CASE 620

FIGURE 1 — BLOCK DIAGRAM



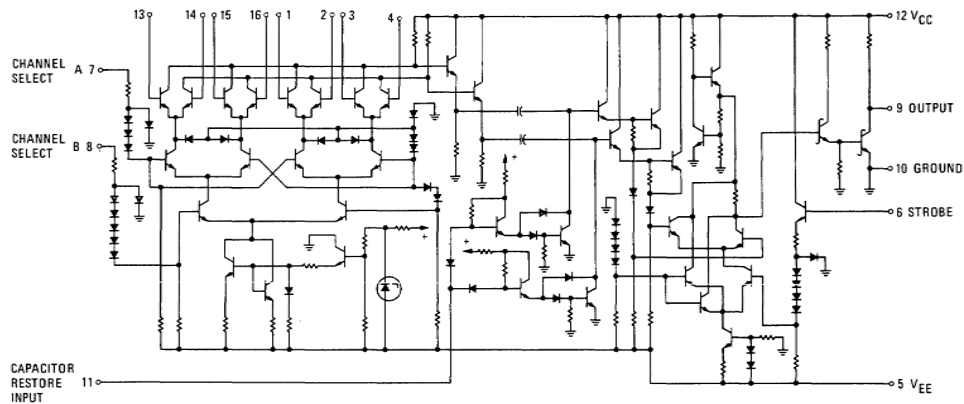
| TRUTH TABLE |       |                  |
|-------------|-------|------------------|
| PIN 7       | PIN 8 | CHANNEL SELECTED |
| HI          | HI    | A                |
| LO          | HI    | B                |
| HI          | LO    | C                |
| LO          | LO    | D                |

# MC1544L, MC1444L (continued)

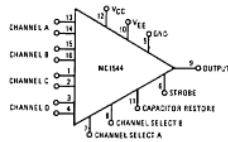
**MAXIMUM RATINGS** ( $T_A = +25^{\circ}\text{C}$  unless otherwise noted)

| RATING                                                                             | SYMBOL                      | VALUE                   | UNIT                        |
|------------------------------------------------------------------------------------|-----------------------------|-------------------------|-----------------------------|
| Power Supply Voltage                                                               | $V_{CC}$<br>$V_{EE}$        | +7.0<br>-8.0            | Vdc                         |
| Common-Mode Input Voltage                                                          | $V_{CM+}$<br>$V_{CM-}$      | +5.0<br>-6.0            | Vdc                         |
| Differential-Mode Input Voltage                                                    | $V_{DM+}$<br>$V_{DM-}$      | +5.0<br>-6.0            | Vdc                         |
| Capacitor Restore, Channel Select, and Strobe Input Voltage                        | $V_{CR}$ , $V_{CS}$ , $V_S$ | +5.5                    | Vdc                         |
| Power Dissipation (Package Limitation)<br>Derate above $T_A = +25^{\circ}\text{C}$ | $P_D$                       | 1.0<br>6.7              | W<br>mW/ $^{\circ}\text{C}$ |
| Operating Temperature Range                                                        | MC1544L<br>MC1444L<br>$T_A$ | -55 to +125<br>0 to +75 | $^{\circ}\text{C}$          |
| Storage Temperature Range                                                          | $T_{stg}$                   | -65 to +150             | $^{\circ}\text{C}$          |
| Junction Temperature                                                               | $T_J$                       | +175                    | $^{\circ}\text{C}$          |

**FIGURE 2 - CIRCUIT SCHEMATIC**



# MC1544L, MC1444L (continued)



**ELECTRICAL CHARACTERISTICS**  
( $T_A = +25^\circ\text{C}$  unless otherwise noted)

| TEST CURRENT/VOLTAGE VALUES |            |             |          |          |          |           |           |           |          |           |          |
|-----------------------------|------------|-------------|----------|----------|----------|-----------|-----------|-----------|----------|-----------|----------|
| $\mu\text{A}$               |            | $\text{mA}$ |          | VOLTS    |          |           |           |           |          |           |          |
| $I_{CM}^+$                  | $I_{CM}^-$ | $I_{OL}$    | $I_{OH}$ | $V_{IL}$ | $V_{IH}$ | $V_{IL2}$ | $V_{IH2}$ | $V_{CC1}$ | $V_{CC}$ | $V_{CC2}$ | $V_{EE}$ |
| 200                         | -10        | 10          | -0.4     | 0.8      | 2.0      | 0         | 3.5       | 4.75      | 5.0      | 5.25      | -5.7     |

| CHARACTERISTIC                            |                | Symbol                                                       | Pin Under Test   | Min    | Typ         | Max    | Unit                               | TEST CURRENT/VOLTAGES APPLIED TO PINS LISTED BELOW: |                |                 |                 |                 |                 |                  |                  |                  |                 |                  |                 |                  |        |          |
|-------------------------------------------|----------------|--------------------------------------------------------------|------------------|--------|-------------|--------|------------------------------------|-----------------------------------------------------|----------------|-----------------|-----------------|-----------------|-----------------|------------------|------------------|------------------|-----------------|------------------|-----------------|------------------|--------|----------|
|                                           |                |                                                              |                  |        |             |        |                                    | I <sub>1</sub>                                      | I <sub>2</sub> | I <sub>OL</sub> | I <sub>OH</sub> | V <sub>IL</sub> | V <sub>IH</sub> | V <sub>IL2</sub> | V <sub>IH2</sub> | V <sub>CC1</sub> | V <sub>CC</sub> | V <sub>CC2</sub> | V <sub>EE</sub> | V <sub>EEH</sub> | GND    |          |
| Input Threshold Voltage (Note 1)          |                | MC1544L                                                      | V <sub>TH</sub>  | 13     | —           | 1.0    | —                                  | mV                                                  | —              | —               | —               | —               | —               | —                | —                | —                | 12              | —                | —               | 5                | —      | 10       |
| T <sub>low</sub> * to T <sub>high</sub> * |                | MC1444L                                                      |                  | 13     | —           | 1.0    | —                                  | mV                                                  | —              | —               | —               | —               | —               | —                | —                | —                | 12              | —                | —               | 5                | —      | 10       |
| Input Bias Current (Note 1)               |                |                                                              | I <sub>B</sub>   | 13     | —           | 20     | —                                  | μA                                                  | —              | —               | —               | —               | —               | 13, 14           | 7, 8             | —                | —               | 12               | —               | —                | 5      | 10       |
| Input Offset Current                      |                |                                                              | I <sub>IO</sub>  | 13, 14 | —           | 1.0    | —                                  | μA                                                  | —              | —               | —               | —               | —               | 13, 14           | 7, 8             | —                | —               | 12               | —               | —                | 5      | 10       |
| Channel Select Input Current (Note 2)     | High Level     | I <sub>CSH</sub>                                             | 7                | —      | 1.8         | 3.0    | mA                                 | —                                                   | —              | —               | —               | —               | —               | —                | 7                | —                | —               | 12               | —               | —                | 5      | 10       |
|                                           | Low Level      | I <sub>CSL</sub>                                             | 7                | —      | 0.6         | 1.0    | mA                                 | —                                                   | —              | —               | —               | —               | —               | 7                | —                | —                | —               | 12               | —               | —                | 5      | 10       |
| Capacitor Restore Input Current           | High Level     | I <sub>CRH</sub>                                             | 11               | —      | 0           | 10     | μA                                 | —                                                   | —              | —               | —               | —               | —               | —                | 11               | —                | —               | 12               | —               | —                | 5      | 10       |
|                                           | Low Level      | I <sub>CRL</sub>                                             | 11               | —      | -2.5        | -3.5   | mA                                 | —                                                   | —              | —               | —               | —               | —               | 11               | —                | —                | —               | 12               | —               | —                | 5      | 10       |
| Strobe Input Current                      | Low/High Level | I <sub>S</sub>                                               | 6                | —      | 40          | 200    | μA                                 | —                                                   | —              | —               | —               | —               | —               | —                | 6                | —                | —               | 12               | —               | —                | 5      | 10       |
| Channel Select Input Voltage (Note 3)     | High Level     | V <sub>CSH</sub>                                             | 7                | 2.1    | 1.6         | —      | V                                  | —                                                   | —              | —               | —               | 7               | 3.8<br>13, 15   | —                | —                | —                | 12              | —                | —               | 5                | —      | 10       |
|                                           | Low Level      | V <sub>CSL</sub>                                             | 7                | —      | 1.2         | 0.7    | V                                  | —                                                   | —              | —               | —               | 7               | 1.8<br>13, 15   | —                | —                | —                | 12              | —                | —               | 5                | —      | 10       |
| Channel Select Input Voltage (Note 3)     | High Level     | V <sub>CSH</sub>                                             | 8                | 2.1    | 1.5         | —      | V                                  | —                                                   | —              | —               | —               | 8               | 1.3<br>7, 13    | —                | —                | —                | 12              | —                | —               | 5                | —      | 10       |
|                                           | Low Level      | V <sub>CSL</sub>                                             | 8                | —      | 1.0         | 0.7    | V                                  | —                                                   | —              | —               | —               | 8               | 1.7<br>13, 15   | —                | —                | —                | 12              | —                | —               | 5                | —      | 10       |
| Capacitor Restore Input Voltage (Note 4)  | High Level     | V <sub>CRH</sub>                                             | 11               | 2.0    | 1.5         | —      | V                                  | —                                                   | —              | —               | —               | 11              | —               | —                | 6                | —                | 12              | —                | —               | 5                | —      | 10       |
|                                           | Low Level      | V <sub>CRL</sub>                                             | 11               | —      | 1.5         | 0.8    | V                                  | —                                                   | —              | —               | —               | 11              | —               | —                | 6                | —                | 12              | —                | —               | 5                | —      | 10       |
| Strobe Input Voltage (Note 4)             | High Level     | V <sub>SH</sub>                                              | 6                | 2.0    | 1.5         | —      | V                                  | —                                                   | —              | —               | —               | 6               | 11              | —                | —                | —                | 12              | —                | —               | 5                | —      | 10       |
|                                           | Low Level      | V <sub>SL</sub>                                              | 6                | —      | 1.5         | 0.8    | V                                  | —                                                   | —              | —               | —               | 6               | 11              | —                | —                | —                | 12              | —                | —               | 5                | —      | 10       |
| Output Voltage                            | High Level     | V <sub>OH</sub>                                              | 9                | 2.4    | 3.6         | —      | V                                  | —                                                   | —              | —               | 9               | 8               | —               | —                | —                | 12               | —               | —                | 5               | —                | 10     |          |
|                                           | Low Level      | V <sub>OL</sub>                                              | 9                | —      | 0.4         | 0.5    | V                                  | —                                                   | —              | —               | 9               | —               | —               | —                | —                | 12               | —               | —                | 5               | —                | 10     |          |
| Power Supply Currents                     | Positive       | I <sub>CC</sub>                                              | 12               | 15     | 22          | 30     | mA                                 | —                                                   | —              | —               | —               | —               | —               | 6, 13, 14        | 7, 8, 11         | —                | —               | 12               | —               | —                | 5      | 10       |
|                                           | Negative       | I <sub>EE</sub>                                              | 5                | 15     | 20          | 30     | mA                                 | —                                                   | —              | —               | —               | —               | —               | 6, 13, 14        | 7, 8, 11         | —                | —               | 12               | —               | —                | 5      | 10       |
| Common-Mode Range Voltage (Note 1)        |                | V <sub>CM</sub> <sup>+</sup><br>V <sub>CM</sub> <sup>-</sup> | 13, 14<br>13, 14 | —<br>— | 4.7<br>-6.0 | —<br>— | V <sub>dc</sub><br>V <sub>dc</sub> | 13, 14<br>13, 14                                    | —              | —               | —               | —               | —               | —                | 7, 8<br>7, 8     | —                | 12<br>12        | —                | —               | 5<br>5           | —<br>— | 10<br>10 |
| Differential-Mode Range Voltage           |                | V <sub>DM</sub>                                              | 13               | —      | 3.7         | —      | V <sub>dc</sub>                    | 13                                                  | —              | —               | —               | —               | —               | 14               | 7, 8             | —                | 12              | —                | —               | 5                | —      | 10       |

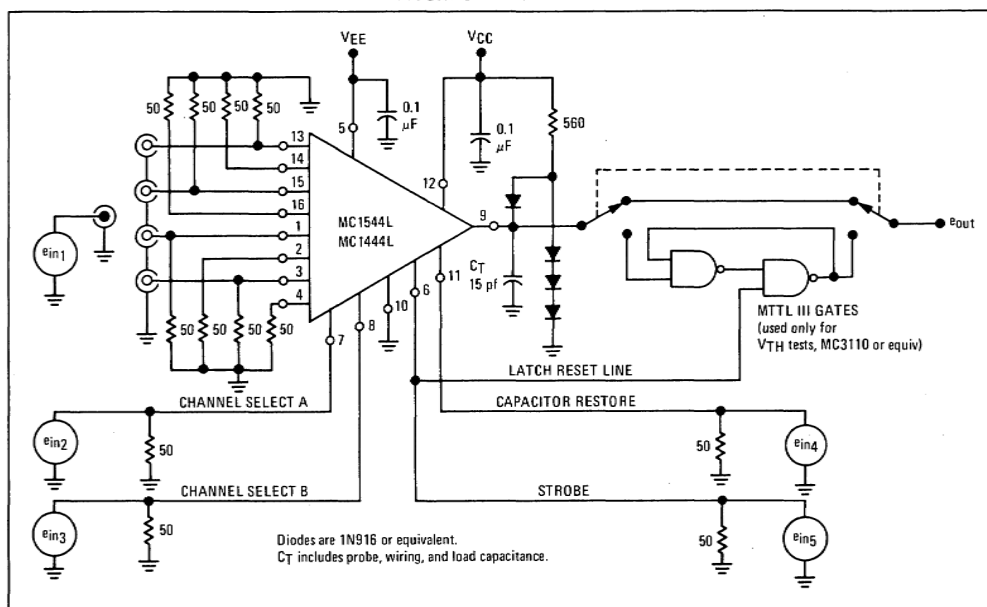
\*MC1544  $T_{low} = -55^\circ\text{C}$ ,  $T_{high} = +125^\circ\text{C}$ ; MC1444  $T_{low} = 0^\circ\text{C}$ ,  $T_{high} = +75^\circ\text{C}$ .

- NOTES: 1. Only one input test is shown; other inputs are tested in the same manner and are selected according to the truth table in Figure 1.  
2. Pin 8 is tested in the same manner.  
3. This requirement is considered satisfied if the input bias currents of all unselected channels total less than 1.0  $\mu\text{A}$  which guarantees that these channels are "off."  
4. This requirement is evaluated during the ac threshold test (Figures 1, 2). A 1.0 mV signal ( $I_{CM1}$ ) is applied to the input,  $V_{CC2}$  will result in  $V_{OH}$  at the output while  $V_{CRL}$  will allow normal operation.  
5. This requirement is evaluated as in Note 4 except  $V_{C2}$  allows normal operation and  $V_{SL}$  causes  $V_{OH}$  at the output.

## SWITCHING CHARACTERISTICS ( $T_A = +25^\circ\text{C}$ unless otherwise noted)

| Characteristic                        | Symbol                     | Figure | Min | Typ      | Max | Unit |
|---------------------------------------|----------------------------|--------|-----|----------|-----|------|
| Propagation Delay Time                | $t_{pd}^+$<br>$t_{pd}^-$   | 1, 5   | -   | 18<br>40 | 25  | ns   |
| Strobe to Input Lead Time             | $t_{SI}$                   | 1, 5   | -   | 10       | -   | ns   |
| Strobe to Output Delay Time           | $t_{SO}^+$<br>$t_{SO}^-$   | 1, 6   | -   | 18<br>30 | 25  | ns   |
| Channel Select to Input Lead Time     | $t_{CSI}$                  | 1, 5   | -   | 15       | -   | ns   |
| Channel Select to Output Delay Time   | $t_{CSO}^+$<br>$t_{CSO}^-$ | 1, 7   | -   | 25<br>40 | -   | ns   |
| Capacitor Restore to Input Lead Time  | $t_{CRI}$                  | 1, 5   | -   | 10       | -   | ns   |
| Capacitor Restore Time (50 mV Offset) | $t_{CR}$                   | 1, 8   | -   | 15       | -   | ns   |
| Common-Mode Recovery Time             | $t_{CRM}^+$<br>$t_{CRM}^-$ | 19     | -   | 50<br>50 | -   | ns   |
| Differential-Mode Recovery Time       | $t_{DMR}^+$<br>$t_{DMR}^-$ | 20     | -   | 65<br>65 | -   | ns   |

FIGURE 3 – AC TEST CIRCUIT



**FIGURE 4 – THRESHOLD VOLTAGE TEST**

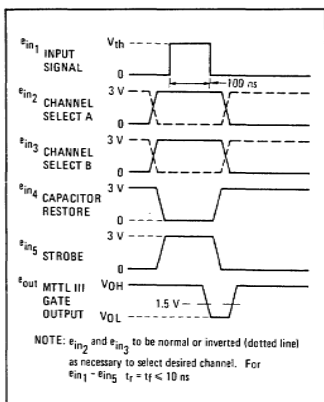


FIGURE 5 –  $t_{csi}$ ,  $t_{cri}$ ,  $t_{si}$ ,  $t_{pd-}$ ,  $t_{pd+}$

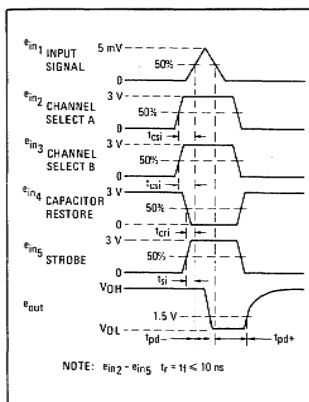


FIGURE 6 –  $t_{SO-}$ ,  $t_{SO+}$

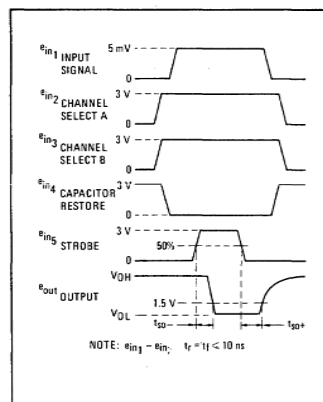
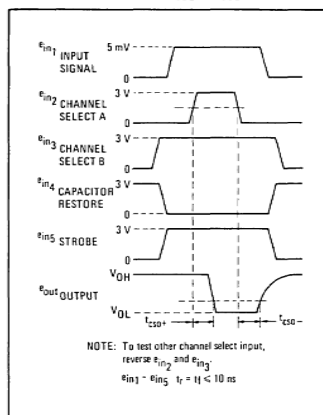
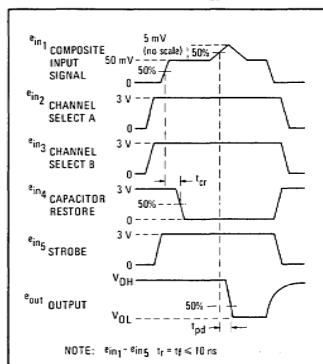


FIGURE 7 —  $t_{cso+}$ ,  $t_{cso-}$ FIGURE 8 —  $t_{cr}$ 

## DEFINITIONS

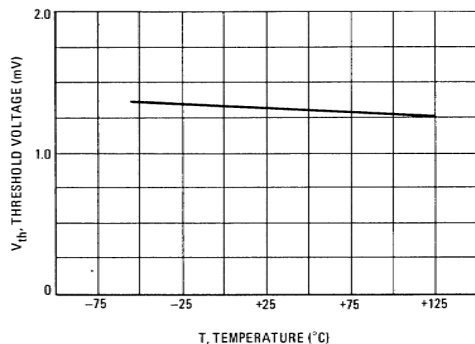
- $I_b$  Input current to the base of any input transistor when the base of the other transistor of the differential pair is at the same voltage
- $I_{CC}$  Positive power supply current
- $I_{CRH}$  The current into the channel select input when the input is at a high-level of 3.5 volts
- $I_{CRL}$  The current out of the capacitor restore input when the input is at a low-level of 0 volts
- $I_{CSH}$  The input current to a channel select input when that input is at a high-level of 3.5 volts
- $I_{CSL}$  The current into a channel select input when the input is at a low-level of 0 volts
- $I_{EE}$  Negative power supply current
- $I_{IO}$  The difference between the base currents of any input differential pair of transistors when the base voltages are equal
- $I_{OH}$  Output logic "1" state source current
- $I_{OL}$  Output logic "0" state sink current
- $I_{SH}$  The current into the strobe input when the input is at a high-level of 3.5 volts
- $I_{SL}$  The current into the strobe input when the input is at a low-level of 0 volts
- $t_{CMR\pm}$  The minimum time between the 50% level of the trailing edge of a + or - 2 volt common-mode signal ( $t_r = t_f \leq 15$  ns) and the 50% level of the leading edge of a 5 mV input pulse when the capacitor restore and strobe inputs are used in a normal manner as shown in Figure 21
- $t_{cr}$  The minimum time between the 50% level of the leading edge of a 50 mV input offset signal and the 50% level of the leading edge of the capacitor restore pulse as shown in Figure 8
- $t_{cri}$  The minimum time between the 50% level of the leading edge of the capacitor restore signal and the 50% level of the leading edge of a 5 mV input signal as shown in Figure 5
- $t_{csi}$  The minimum time between the 50% level of the leading edge of the channel select and the 50% level of the leading edge of a 5 mV input signal as shown in Figure 5
- $t_{cso+}$  The delay time from the 50% level of the trailing edge of the channel select signal to the 1.5 volt level of the positive edge of the output when the input to the selected channel is held at the "1" level as shown in Figure 7
- $t_{cso-}$  The delay time from the 50% level of the leading edge of the channel select signal to the 1.5 volt level of the negative edge of the output when the input to the selected channel is held at the "1" level as shown in Figure 7
- $t_{DMR\pm}$  The minimum time between the 50% level of the trailing edge of a + or - 1 volt differential-mode signal ( $t_r = t_f \leq 15$  ns) and the 50% level of the leading edge of a 5 mV input pulse when the capacitor restore and strobe inputs are used in a normal manner as shown in Figure 22
- $t_{pd+}$  The delay time from the 50% level of the trailing edge of a 5 mV input signal to the 1.5 volt level of the positive edge of the output as shown in Figure 5
- $t_{pd-}$  The delay time from the 50% level of the leading edge of a 5 mV input signal to the 1.5 volt level of the negative edge of the output as shown in Figure 5
- $t_{si}$  The minimum time between the 50% level of the leading edge of the strobe and the 50% level of the leading edge of the input signal as shown in Figure 5
- $t_{so+}$  The delay time from the 50% level of the trailing edge of the strobe to the 1.5 volt level of the positive edge of the output when the input is held at the "1" level as shown in Figure 6
- $t_{so-}$  The delay time from the 50% level of the leading edge of the strobe to the 1.5 volt level of the negative edge of the output when the input is held at the "1" level as shown in Figure 6
- $V_{CC}$  Positive power supply voltage
- $V_{CCH}$  Maximum operating positive power supply voltage
- $V_{CCL}$  Minimum operating positive power supply voltage
- $V_{CM+}$  The maximum common-mode input voltage that will not saturate the amplifier
- $V_{CM-}$  The minimum common-mode input voltage that will not break down the amplifier
- $V_{CRH}$  The minimum high-level voltage at the capacitor restore input required to insure that the capacitors are clamped i.e., the input threshold voltage is greater than 10 mV
- $V_{CRL}$  The maximum low-level voltage at the capacitor restore input which will allow normal operation during the threshold test
- $V_{CSH}$  The minimum high-level voltage at a channel select input required to insure that the total of the base currents of all unselected inputs is less than  $1.0 \mu A$
- $V_{CSL}$  The maximum low-level voltage at a channel select input required to insure that the total of the base currents of all unselected inputs is less than  $1.0 \mu A$
- $V_{DM}$  The maximum differential-mode input voltage that will not saturate the amplifier
- $V_{EE}$  Negative power supply voltage
- $V_{EEH}$  Maximum operating negative power supply voltage
- $V_{EEL}$  Minimum operating negative power supply voltage
- $V_{OH}$  Logic "1" state output voltage
- $V_{OL}$  Logic "0" state output voltage
- $V_{SH}$  The minimum high-level voltage at the strobe input which will allow normal operation during the threshold test
- $V_{SL}$  The maximum low-level voltage at the strobe input which will result in  $V_{OH}$  at the output regardless of input signals
- $V_{th}$  The minimum input signal ( $e_{in1}$ ) required to drive the MTTL III gates to obtain the  $e_o$  waveform shown in Figure 4



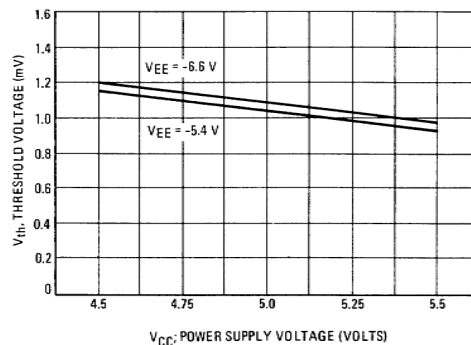
**TYPICAL CHARACTERISTICS**

( $T_A = +25^\circ\text{C}$  unless otherwise noted)

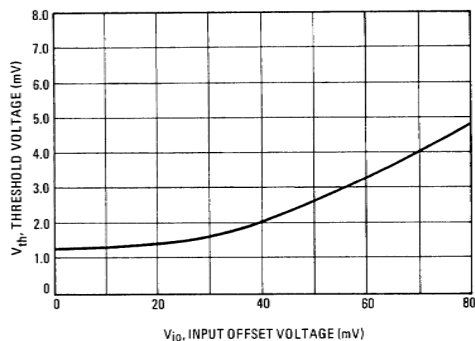
**FIGURE 9 – THRESHOLD VOLTAGE versus TEMPERATURE**



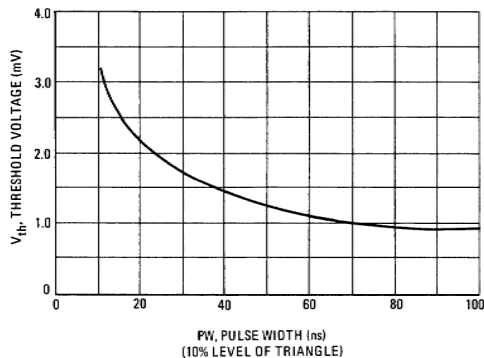
**FIGURE 10 – THRESHOLD VOLTAGE versus POWER SUPPLIES**



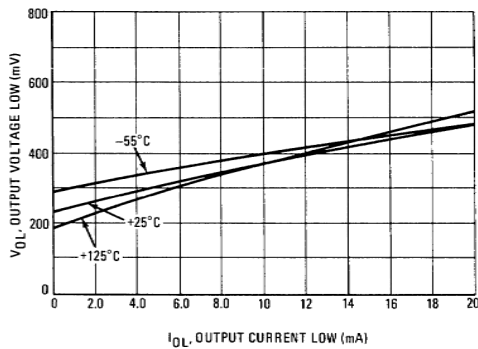
**FIGURE 11 – THRESHOLD versus INPUT OFFSET VOLTAGE**



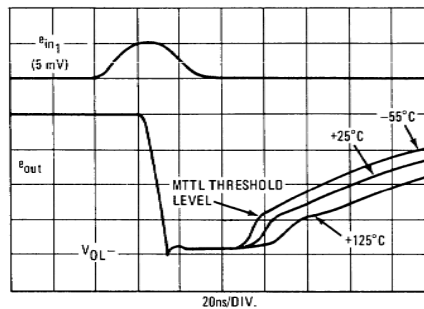
**FIGURE 12 – THRESHOLD VOLTAGE versus PULSE WIDTH**



**FIGURE 13 – OUTPUT VOLTAGE versus CURRENT and TEMPERATURE**



**FIGURE 14 – SENSE AMPLIFIER RESPONSE versus TEMPERATURE (See Figures 3 and 5)**



TYPICAL CHARACTERISTICS (continued)

FIGURE 15 – INPUT IMPEDANCE versus FREQUENCY

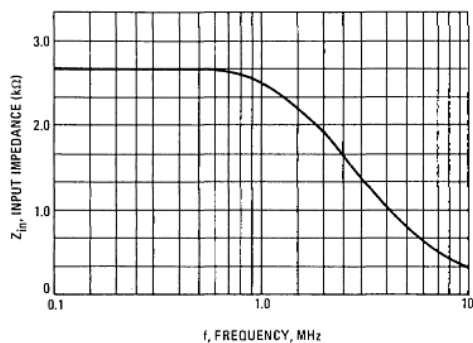


FIGURE 16 – CAPACITOR RESTORE TIME versus INPUT OFFSET VOLTAGE

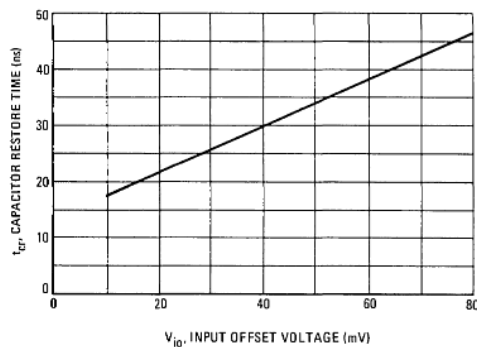


FIGURE 17 – AMPLIFIER INPUT TO OUTPUT TRANSFER CHARACTERISTIC

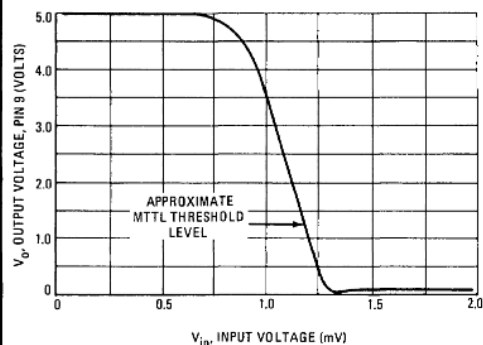


FIGURE 18 – STROBE TO OUTPUT TRANSFER CHARACTERISTICS

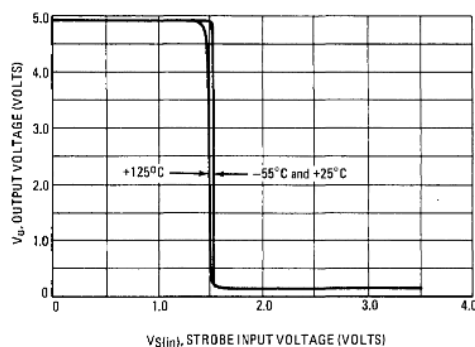


FIGURE 19 – CHANNEL SELECT A to OUTPUT TRANSFER CHARACTERISTICS

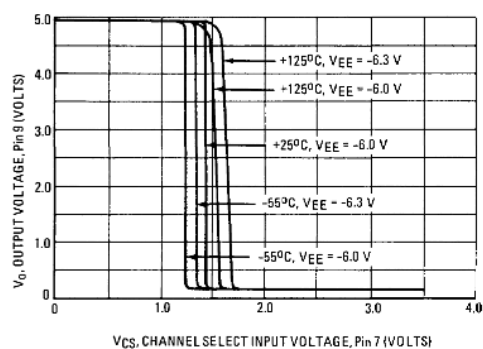


FIGURE 20 – CHANNEL SELECT B to OUTPUT TRANSFER CHARACTERISTICS

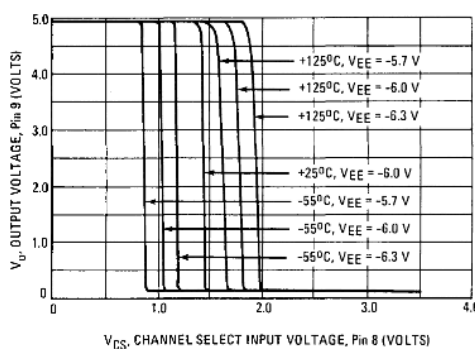


FIGURE 21 – COMMON-MODE CHARACTERISTICS

Note: The 5mV Input Signal (Differential) is superimposed on the Common-Mode Input and is shown separately for reference only.

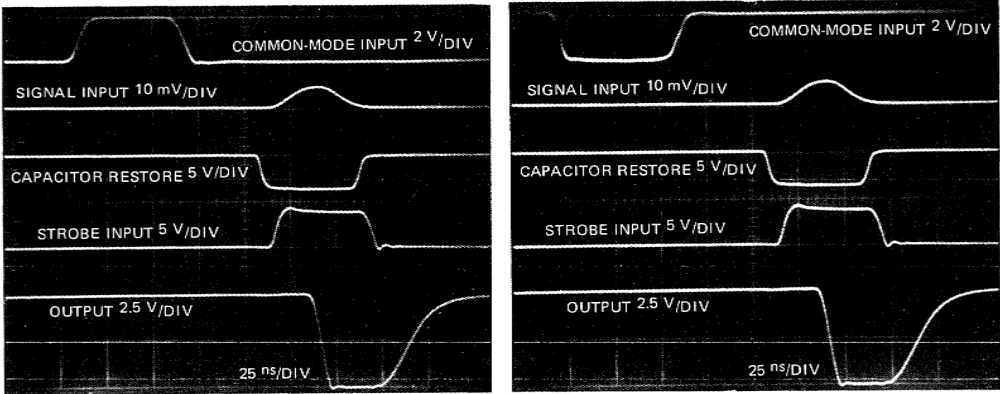


FIGURE 22 – DIFFERENTIAL-MODE CHARACTERISTICS

Note: The 5mV Input Signal is superimposed on the Differential Input and is shown separately for reference only.

